Lecture 1 : Introduction to optimization and refresher course

1 Course logistics

Below are three main points of this course:

1. **Calendar**: 4h30 per week, in which:
   - 1 TP at 9h45 AM in Monday morninng (IM2AG F202).
   - 1 lecture and 1 TD from 9h45 AM to 1h PM on Thursday (IM2AG F321).
2. **Main objectives**: This course aims to cover the most basic elements of numerical (continuous) optimization. Notable components are:
   - Theory of unconstrained and constrained optimization.
   - Convex optimization.
   - Optimization algorithms: (stochastic) gradient descent, acceleration, second-order methods, with theoretical guarantees.
   - Nonsmooth optimization, proximal operators.
3. **Evaluation**: one midterm (1h) and one final exam (2h). Both are written exams. Their contribution are 0.3 and 0.7.

2 Several examples of optimization problems

We consider three examples:

Linear regression We have a statistical serie $\{(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R} \mid i = 1, \ldots, n\}$. We suppose that the relation between x and y is linear, i.e., $y \approx \theta^\top x$, find the best θ to fit the given serie.

$$\underset{\theta \in \mathbb{R}^d}{\text{Minimize}} \quad \mathcal{L}(\theta) = \frac{1}{2n} \sum_{i=1}^{n} (x_i^\top \theta - y_i)^2 = \frac{1}{2n} \|\mathbf{X}\theta - y\|_2^2, \tag{1}$$

where

$$\mathbf{X} = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix} \in \mathbb{R}^{n \times d}, \quad y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n, \quad \|v\|_2^2 = \sum_{i=1}^{n} v_i^2, v \in \mathbb{R}^n.$$

Image denoising We have access to a noisy version of an image $\tilde{X} \in [0,1]^{d\times d}$ (assuming that the image is black and white). We would like to denoise $\tilde{X}$. We use a simple observation: in a natural image, close pixels tend to have similar values. Therefore, we can consider the following problems:

$$\underset{X\in[0,1]^{d\times d}}{\text{Minimize}} \quad \mathcal{L}(\mathbf{X}) = \|\mathbf{X} - \tilde{\mathbf{X}}\|_F^2 + \lambda \sum_{i,j} \sum_{(i',j')\in\mathcal{N}(i,j)} |\mathbf{X}_{i,j} - \mathbf{X}_{i',j'}| \tag{2}$$

where:

$$\|\mathbf{X}\|_F^2 = \sum_{i,j} \mathbf{X}_{i,j}^2$$

and $\mathcal{N}(i,j)$ is the set of pixels neighboring the pixel (i,j).

Diet problem - Linear programming We have n types of food, each of which provides m types of nutritions. In detail, each unit of the ith food provides $\mathbf{A}_{i,j}$ gram of the jth nutrition. A normal person needs b_j gram of the jth nutrition to live healthily. The ith food costs c_i euros per unit. Find the cheapest strategy for a man to have enough nutrition.

$$\begin{aligned} \underset{x\in\mathbb{R}^n}{\text{Minimize}} \quad & c^\top x \\ \text{subject to} \quad & \mathbf{A}x \geq b \\ & x \geq 0 \end{aligned}$$

The answer: cabbage, spinach, wheat flour, evaporate milk, beans (Stigler diet).

3 Refresher elements: gradient, hessian matrix, Taylor expansion, properties of optimal solutions

Definition 3.1 (Derivatives of a function)**.** A function $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable at a point x if there exists a vector $\nabla f(x)$ such that:

$$\lim_{d\to 0} \frac{\|f(x+d) - f(x) - \langle \nabla f(x), d\rangle\|}{\|d\|} = 0,$$

or equivalently,

$$f(x+d) = f(x) + \langle \nabla f(x), d\rangle + r_f(d) \quad \text{where } \lim_{d\to 0} \frac{\|r_f(d)\|}{\|d\|} = 0.$$

The vector $\nabla f(x)$ is called the gradient of f at $x \in \mathbb{R}$. Moreover, the vector $\nabla f(x)$ is given by:

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_d}(x) \end{pmatrix} \in \mathbb{R}^d,$$

where each coordinate is given by the derivatives of the function $g_i : \mathbb{R} \to \mathbb{R} : x_i \to f(x_1, \ldots, x_i, \ldots, x_n)$, or equivalently:

$$\frac{\partial f}{\partial x_i}(x) = g_i'(x_i) = \lim_{t\to 0} \frac{f(x_1, \ldots, x_i + t, \ldots, x_d) - f(x_1, \ldots, x_i, \ldots, x_n)}{t}.$$

Remark 3.2. If f is differentiable at x, then f has to be continuous at x.

Example 3.3. For example, $f(x) = \|x\|_2^2$ is differentiable at $x = 0$, but the function $f(x) = |x|$ is not.

Definition 3.4 (Differentiable and continuously differentiable (C^1) functions)**.** If a function $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable at all point $x \in \mathbb{R}^d$, then f is differentiable. If the mapping $x \mapsto \nabla f(x) : \mathbb{R}^d \to \mathbb{R}^d$ is continuous, then we say that f is continuously differentiable, or a C^1 function.

Proposition 3.5 (Properties of derivatives and gradient)**.** *Given two differentiable functions $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$, we have:*

$$\begin{aligned}
\nabla(f+g)(x) &= \nabla f(x) + \nabla g(x) \\
\nabla(\alpha f)(x) &= \alpha \nabla f(x), \forall \alpha > 0 \\
\nabla(f \cdot g)(x) &= g(x)\nabla f(x) + f(x)\nabla f(x) \\
\nabla\left(\frac{f}{g}\right) &= \frac{g(x)\nabla f(x) - f(x)\nabla g(x)}{g(x)^2}, \quad \textit{assuming that } g(x) > 0.
\end{aligned} \tag{3}$$

Proof. Proof is left as exercise. □

Definition 3.6 (Generalization for vector-valued function)**.** A function $f : \mathbb{R}^d \to \mathbb{R}^\ell : x \mapsto (f_1(x), \dots, f_\ell(x))$ is differentiable at a point x if $f_i(x)$ is differentiable at x for all $i = 1, \dots, \ell$. In that case, the derivatives of f at point x given by a matrix (known as Jacobian matrix) whose formulation is:

$$J_f(x) = \begin{pmatrix} \nabla f_1(x)^\top \\ \vdots \\ \nabla f_\ell(x)^\top \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) & \cdots & \frac{\partial f_1}{\partial x_d}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_\ell}{\partial x_1}(x) & \cdots & \frac{\partial f_\ell}{\partial x_d}(x) \end{pmatrix}.$$

We also have:

$$\lim_{d \to 0} \frac{\|f(x+d) - f(x) - J_f(x)d\|}{\|d\|} = 0.$$

Example 3.7. Consider a vector-valued differentiable function such as $f : \mathbb{R}^2 \to \mathbb{R}^2 : (x, y) \to (x^2y + x^2, y^3)$.

Proposition 3.8 (Chain rule)**.** *Given two differentiable functions $f : \mathbb{R}^k \to \mathbb{R}^\ell$ and $g : \mathbb{R}^d \to \mathbb{R}^k$, then the composition $f \circ g : \mathbb{R}^d \to \mathbb{R}^\ell$ is also differentiable and its Jacobian matrix is given by:*

$$J_{f \circ g}(x) = J_f(g(x))J_g(x).$$

Proof. Proof is left as exercise. □

Definition 3.9 (Twice differentiable functions and Hessian matrix)**.** A function $f : \mathbb{R}^d \to \mathbb{R}^d$ is called twice differentiable at a point x if the function $x \to \nabla f(x) : \mathbb{R}^d \to \mathbb{R}^d$ is differentiable (in the sense of a vector-valued function). The hessian matrix is the Jacobian matrix of $x \to \nabla f(x)$, i.e.,

$$H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(x) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1}(x) & \cdots & \frac{\partial^2 f}{\partial x_d \partial x_d}(x) \end{pmatrix}$$

where coefficients are given by:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_j}\right)(x).$$

f is twice differentiable if f is twice differentiable at all $x \in \mathbb{R}^d$.

f is twice continuously differentiable (or a C^2 function) if the mapping $x \to H_f(x)$ is continuous. In that case, $H_f(x)$ is symmetric for all $x \in \mathbb{R}^d$.

Example 3.10. Consider:

1. $f(x) = \frac{1}{2}\|x\|_2^2$, we have $\nabla f(x) = x$ and $H_f(x) = \mathbf{I}$.
2. $f(x) = \sum_{i=1}^d x_i$, we have $\nabla f(x) = x$ and $H_f(x) = 0$.
3. $f(x) = \frac{1}{2}(\sum_{i=1}^d x_i)^2$, we have $\nabla f(x) = (\sum_{i=1}^k x_i)\mathbf{1}_d$ and $H_f(x) = \mathbf{1}_{d\times d}$.

Proposition 3.11 (Two Taylor formulations)**.** *Given a C^1 (resp. C^2) function $f : \mathbb{R}^d \to \mathbb{R}$, we have:*

$$f(y) = f(x) + \int_0^1 \nabla f(x + t(y-x))^\top (y-x)\, dt \qquad ,\forall x, y \in \mathbb{R}^d$$

$$(resp.) f(y) = f(x) + (y-x)^\top \nabla f(x) + \frac{1}{2}(y-x)^\top \nabla^2 f(x)(y-x) + R_2(x-y) \quad ,\forall x, y \in \mathbb{R}^d, \tag{4}$$

where $R_2(x-y)$ is a reminder satisfying $\lim_{y\to x} \dfrac{R_2(x-y)}{\|y-x\|^2} = 0$.

Proof. Proof is left as an exercise. □

3.1 Optimal solutions of an optimization problem and their properties

In this section, we consider the following optimization problem:

$$\underset{x\in\mathcal{F}}{\text{Minimize}} \quad f(x), \quad \mathcal{F} \subseteq \mathbb{R}^d. \tag{OP}$$

where $\mathcal{F}$ is called the feasible set and its elements are called admissible points. In the following, we remind the definition of an optimal solution of (OP).

Definition 3.12 (Global solutions)**.** A point $x^\star \in \mathcal{F}$ is called an global optimal solution of (OP) if $f(x^\star)$ is the smallest value of f in $\mathcal{F}$, i.e.:

$$f(x^\star) \leq f(x), \qquad \forall x \in \mathcal{F}.$$

In general, optimal solutions of an optimization problem are difficult to find (unless under certain hypothesis such as convexity). Therefore, we will introduce another notion of optimal solutions, which is more accessible in practice:

Definition 3.13 (Local solutions)**.** A point $x^\star \in \mathcal{F}$ is called an local optimal solution of (OP) if $f(x^\star)$ is the smallest value of f in a admissible neighborhood of $x^\star$, i.e., there exists $\delta > 0$ such that:

$$f(x^\star) \leq f(x), \qquad \forall x \in B_\delta(x^\star) \cap \mathcal{F}.$$

where $B_\delta(x^\star)$ is the open ball centered at $x^\star$ and of radius δ.

It is clear that a global solution is also local. However, a local solution is not necessarily global. In the case where $\mathcal{F} = \mathbb{R}^d$ (unconstrained optimization), we introduce a necessary condition so that a point $x^\star$ is an optimal solution of (OP).

Theorem 3.14 (Necessary conditions)**.** *Consider* (OP) *where $\mathcal{F} = \mathbb{R}^d$. If a point $x^\star$ is a local solution of* (OP)*, then:*

1. *If f is C^1, $\nabla f(x^\star) = 0$.*
2. *If f is C^2, we have in addition that $\nabla^2 f(x^\star) \succeq 0$, i.e., $v^\top \nabla^2 f(x^\star) v \geq 0, \forall v \in \mathbb{R}^d$.*

Proof. Both items are proved using (4):

1. Proof of $\nabla f(x^\star) = 0$: By contradiction, we assume that $d := \nabla f(x^\star) \neq 0$. We consider $y = x^\star - ad$ where a is a constant to be determined. By Taylor formula, we have:

$$f(y) = f(x^\star) - \underbrace{a \int_0^1 d^\top \nabla f(x^\star - tad) dt}_{C}.$$

Since f is C^1 and $d^\top \nabla f(x^\star) = \|d\|_2 > 2$, for all sufficiently small $a > 0$, we have: $d^\top \nabla f(x^\star - tad) \geq c > 0$. By consequent,

$$f(y) \leq f(x^\star) - ac < f(x^\star)$$

for all a sufficiently small. This is a contradiction to the assumption that $x^\star$ is a local solution.

2. Proof of $\nabla^2 f(x^\star) \succeq 0$: Assume that $\nabla^2 f(x^\star)$ is not semi-definite positive. In that case, there exists a vector $d \in \mathbb{R}^d$ such that $c := d^\top \nabla^2 f d < 0$ and $\|d\| = 1$. By Taylor formulation:

$$\begin{aligned} f(x^\star - td) &= f(x^\star) + t\nabla f(x^\star)^\top d + \frac{1}{2} t^2 d^\top \nabla^2 f(x^\star) d + R_2(td) \\ &= f(x^\star) + \frac{t^2}{2} c + R_2(td). \end{aligned}$$

Since $\lim_{t \to 0} \frac{R_2(td)}{t^2} = 0$, for all t sufficiently small, we have $f(x^\star - td) < f(x^\star)$. This is also a contraction to $x^\star$ being a local solution.

□

Remark 3.15. The conditions Theorem 3.14 are only necessary and not sufficient. Take $f(x) = \frac{1}{6}x^3$ and $x^\star = 0$. Although $\nabla f(x^\star) = \nabla^2 f(x^\star) = 0$, $x^\star$ is neither a global, nor a local solution of f.

Definition 3.16. A point $x^\star \in \mathcal{F}$ is called a critical point of a function C^1 f if $\nabla f(x^\star) = 0$.

We have the following relations:

$$x^\star \text{ is a global solution} \implies x^\star \text{ is a local solution} \implies x^\star \text{ is a critical point.}$$

The converse is not true in general.

Theorem 3.17 (Sufficient condition)**.** *Consider* (OP) *where $\mathcal{F} = \mathbb{R}^d$ and f is a C^2 function. If a point $x^\star$ satisfies :*

1. $\nabla f(x^\star) = 0$.
2. $\nabla^2 f(x^\star) \succ 0$ *(i.e., $\forall v \in \mathbb{R}^d, \|v\| = 1$, $v^\top \nabla^2 f(x^\star) v > 0$),*

then $x^\star$ is a local solution of (OP)*.*

Proof. Take $c := \min_{\|v\|=1} v^\top \nabla^2 f(x^\star) v > 0$. Therefore, $\forall \delta \in \mathbb{R}$, we have $\delta^\top \nabla^2 f(x^\star) \delta \geq c\|\delta\|^2$. The Taylor formulation gives us :

$$\begin{aligned} f(x^\star + \delta) &= f(x^\star) + \nabla f(x^\star)^\top \delta + \frac{1}{2} \delta^\top \nabla^2 f(x^\star) \delta + R_2(\delta) \\ &= f(x^\star) + \frac{1}{2} \delta^\top \nabla^2 f(x^\star) \delta + R_2(\delta) \\ &\geq f(x^\star) + \frac{c}{2} \|\delta\|^2 + R_2(\delta). \end{aligned}$$

Since $\lim_{\delta \to 0} \frac{R_2(\delta)}{\|\delta\|^2} = 0$, there exists a sufficiently small neighborhood of $x^\star$ where $f(x) > f(x^\star)$. Thus, $x^\star$ is a local solution of (OP). □

Lecture 2 : Convex functions and sets

1 Introduction

So far, we already saw that it is necessary for a (local/global) solution $x^\star$ to be a critical point, i.e., $\nabla f(x^\star) = 0$. In general, the set of critical points of an optimization problem is strictly larger than the set of local/global minima.

In this lecture, we will investigate a family of optimiation problems where these two sets coincide. It is known as convex optimization. We will proceed two main notions in this lecture: convex sets and convex functions.

2 Convex sets

Definition 2.1 (Convex sets)**.** A set A is convex if for all elements $x, y \in A$, we have:

$$tx + (1-t)y \in A, \qquad \forall t \in [0,1]$$

Example 2.2. Below are several examples of convex (and nonconvex) sets:

1. The empty set $\emptyset$, singleton $\{x\}, x \in \mathbb{R}^d$ and the whole space $\mathbb{R}^d$ are convex.
2. Any hyperplane of the form $\{x \mid a^\top x = b\} \subseteq \mathbb{R}^d$ is convex.
3. Any halfspace of the form $\{x \mid a^\top x \leq b\} \subseteq \mathbb{R}^d$ is convex.
4. The sphere $\{x \mid \|x\|^2 = 1\} \subseteq \mathbb{R}^d$ is not convex.

Definition 2.3 (Convex hull of a set)**.** The convex hull of a set A is the set of all convex combination of finite points of A, i.e.,

$$\mathbf{conv}(A) = \{\theta_1 x_1 + \ldots + \theta_k x_k \mid x_i \in C, \theta_i \geq 0, \sum_{i=1}^{k} \theta_k = 1\}.$$

Proposition 2.4 (Convex hull of a set is convex)**.** *The convex hull of a set A is the smallest convex set containing A.*

Proof. For any pair $(x_1, x_2) \in \mathbf{conv}(A) \times \mathbf{conv}(A)$, we want to prove that:

$$tx_1 + (1-t)x_2 \in \mathbf{conv}(A), \quad \forall t \in [0,1].$$

WLOG, one can assume that:

$$x_i = \sum_{j=1}^{k} \theta_{i,j} a_j, \qquad \theta_{i,j} \geq 0, \forall 1 \leq j \leq k, \quad \sum_{j=1}^{k} \theta_{i,j} = 1.$$

Then,

$$\begin{aligned} tx_1 + (1-t)x_2 &= t\left(\sum_{j=1}^{k} \theta_{1,j} a_j\right) + (1-t)\left(\sum_{j=1}^{k} \theta_{2,j} a_j\right) \\ &= \sum_{j=1}^{k} \underbrace{(t\theta_{1,j} + (1-t)\theta_{2,j})}_{\theta_j} a_j \end{aligned}$$

where $\theta_j \geq 0$ and $\sum_{j=1}^{k} \theta_j = t(\sum_{j=1}^{k} \theta_{1,j}) + (1-t)(\sum_{j=1}^{k} \theta_{2,j}) = t + (1-t) = 1$. By definition, $tx_1 + (1-t)x_2$ is an element of $\mathbf{conv}(A)$. Therefore, $\mathbf{conv}(A)$ is convex.

The next claim is proved by using the following result (which will be discussed in the tutorial session).

Lemma 2.5 (Convex combination)**.** *If A is a convex set, then any convex combination of finite points of A is an element of A.*

To this end, assume that B to be a convex set that contains A. By the previous lemma, we have: $\mathbf{conv}(A) \subseteq B$. Hence, the claim is proved. □

Example 2.6. Using Proposition 2.4, we can construct even more convex sets.

1. The hypercube $\{x \mid 0 \leq x_i \leq 1\}$ is the convex hull of $A = \{0,1\}^d$.
2. The ball $\{x \mid \|x\| \leq 1\}$ is the convex hull of the sphere of the same dimension.
3. The probability simplex $\{x \mid x_i \geq 0, \sum_i x_i = 1\}$ is the convex hull of $A = \{e_i \mid i = 1, \ldots, d\} \cup \{0\}$.

Proposition 2.7 (Operation preserving convexity)**.** *The following operations preserve convexity:*

1. *(Infinite) intersection of convex sets is convex.*
2. *Affine transformation of a convex set A, i.e., a set of the form $\{\mathbf{A}x + b \mid x \in A\}$, is convex.*

Proof. We prove two claims one by one:

1. Consider $S = \cup_{i \in \mathcal{I}} S_i$ where $S_i, i \in \mathcal{I}$ are convex sets. For any pair $(x, y) \in S \times S$, we also have $(x, y) \in S_i \times S_i, \forall i \in \mathcal{I}$. Since S_i is convex (by assumption), we have:

$$tx + (1-t)y \in S_i, \forall i \in \mathcal{I}, \forall t \in [0,1].$$

   Therefore, $tx + (1-t)y \in S$ and S is convex.

2. By the fact that $t(\mathbf{A}x + b) + (1-t)(\mathbf{A}y + b) = \mathbf{A}(tx + (1-t)y) + b$.

□

Example 2.8. We revisit previous examples and prove their convexity by Proposition 2.7:

1. The hypercube $\{x \mid 0 \leq x_i \leq 1\}$ is the intersection of the half spaces $x_i \geq 0$ and $x_i \leq 1, i = 1, \ldots, d$.

2. The ball $\{x \mid \|x\| \leq 1\}$ is the intersection of the the half spaces $\{x \mid x^\top d \geq 1\}, d \in \mathbb{R}^d, \|d\| = 1$.
3. The probability simplex $\{x \mid x_i \geq 0, \sum_i x_i = 1\}$ is the intersection of the half spaces $x_i \geq 0$ and $\sum_i x_i \leq 1$.
4. The projection of a convex set to some linear subspaces is convex.
5. Scaling or translation of a convex set is convex.

Remark 2.9. In fact, the examples demonstrates the following result (not proved): any closed convex set is equal to the intersection of (infinite) half spaces.

3 Convex functions

Definition 3.1 (Convex functions)**.** A function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is convex if its domain A is convex and for all $x, y \in A$, we have:

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y), \quad \forall t \in [0,1].$$

Example 3.2. Below are examples of convex functions:

1. Affine function: $f(x) = a^\top x + b$.
2. Any norm function: $f(x) = \|x\|$.
3. ReLU: $f : \mathbb{R} \to \mathbb{R} : x \mapsto \max(x, 0)$.

In the following, we present three equivalent definitions of convex functions $f : \mathbb{R}^d \to \mathbb{R}$.

Definition 3.3 (Equivalent definitions of convex functions)**.** Three following conditions are equivalent:

1. $f : \mathbb{R}^d \to \mathbb{R}$ is convex.
2. If f is C^1: $f(y) \geq f(x) + \nabla f(x)^\top (y - x), \forall x, y \in \mathbb{R}^d$.
3. If f is C^2: $\nabla^2 f(x) \succeq 0$.

Proof. We prove that 1 $\iff$ 2 and 2 $\iff$ 3.

1. If f is C^1: Since $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$, we have:

$$\frac{f(x + t(y-x)) - f(x)}{t} \leq f(y) - f(x), \forall x, y \in \mathbb{R}, \forall t \in (0,1].$$

Taking the limit of the LHS, we obtain:

$$\nabla f(x)^\top (y - x) \leq f(y) - f(x).$$

The converse can be proved by taking the weighted sum of the two following equations:

$$f(x) \geq f(tx + (1-t)y) + (1-t)\nabla f(tx + (1-t)y)^\top (x - y).$$
$$f(y) \geq f(tx + (1-t)y) - t\nabla f(tx + (1-t)y)^\top (x - y).$$

2. If f is C^2: By contradiction, we suppose that $\nabla^2 f(x)$ is not positive semidefinite. By consequence, there exists a vector d of norm 1 such that:

$$c := d^\top \nabla^2 f(x) d < 0.$$

We consider the Taylor formula in a neighborhood of x, we get:

$$f(\underbrace{x+td}_{y}) = f(x) + \nabla f(x)^\top (y-x) + t^2 \underbrace{d^\top \nabla^2 f x) d}_{c} + o(t^2)$$

This equation shows that for sufficiently small t, we have: $f(y) < f(x) + \nabla f(x)^\top (y-x)$, which contradicts the second characterization of the convexity of f.

The converse can be proved by using an alternative form of the Taylor: Given a function $g : \mathbb{R} \to \mathbb{R}$ of class C^2 and two points $x, y \in \mathbb{R}$, there exists $z \in [x, y]$ such that:

$$g(y) = g(x) + g'(x)(y-x) + \frac{1}{2} g''(z)(y-x)^2.$$

In the general case $\mathbb{R}^d$, for two points $x, y \in \mathbb{R}^d$, we consider the following C^2 function:

$$g(t) = f(ty + (1-t)x) = f(x + t(y-x)).$$

By applying the previous form, we obtain:

$$\begin{aligned} f(y) = g(1) &= g(0) + g'(0) + \frac{1}{2} g''(z) \qquad \text{for certain } z \in [0,1] \\ &= f(x) + \nabla f(x)^\top (y-x) + \frac{1}{2}(y-x)^\top \nabla f(z)(y-x) \\ &\geq f(x) + \nabla f(x)^\top (y-x). \end{aligned}$$

□

Example 3.4. All these properties can be used to verfiy the convexity of a function. Consider examples such as $f(x) = \|x\|_2^2$ or $f(x) = \|Ax - b\|_2^2$.

Theorem 3.5 (Optimial conditions of a convex function)**.** *Consider the unconstrained optimization problem:*

$$\underset{x \in \mathbb{R}^d}{\text{Minimize}} \quad f(x)$$

where f is C^1 and convex. A point x is a global solution of $f(x)$ if and only if $\nabla f(x) = 0$, i.e., x is a critical point of f.

Proof. If x is a global solution, then x has to be a critical point of f.

Conversely, if $\nabla f(x) = 0$, by applying the second characterization of the convexity of f, we have:

$$f(y) \geq f(x) + \nabla f(x)^\top (y-x) = f(x), \forall y \in \mathbb{R}^d,$$

which shows that x is indeed a global solution of f. □

Remark 3.6. A related notion of convexity is concavity. A function f is concave if $-f$ is convex. People usually consider minimizing a convex function and maximizing a concave one.

Lemma 3.7 (Operation preserving the convexity)**.** *The following operations preserve the convexity:*

1. *If f is convex, then αf is also convex ($\alpha > 0$).*

2. *If f and g are convex, then $f + g$ is also convex.*
3. *If $f_i, i = 1, \ldots, n$ are convex and $\alpha_i \geq 0, \forall i = 1, \ldots, n$, then $\sum_i \alpha_i f_i$ is also convex.*
4. *If f, g are convex, then $\max(f, g)$ is convex.*
5. *If f is convex, then $g(x) = f(Ax + b)$ is also convex.*
6. *if $f : \mathbb{R}^d \to \mathbb{R}$ is convex and $g : \mathbb{R} \to \mathbb{R}$ is non-decreasing and convex, then $g \circ f$ is convex.*

Lecture 3 : Gradient descent and theoretical properties

1 Introduction

In previous lectures, we already discussed about optimization and several theoretical properties. In this lecture, we will focus on algorithms, which are used to find global/local minima or critical points of optimization algorithm. Our spotlight is gradient descent, one of the most famous optimization algorithms and it is still very useful up to nowsaday standard.

2 Gradient descent algorithm

Consider an unconstrained optimization problem:

$$\underset{x\in\mathbb{R}^d}{\text{Minimize}} \quad f(x) \tag{OP}$$

where f is a C^1 function. Gradient descent search for a solution of (OP) by an iterative procedure, where in each iteration, we update the iterates by:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k). \tag{GD}$$

where $\alpha_k > 0$ is called step-size (also known as learning rate in Machine Learning). We can make several remark about (GD):

1. If $\|x_k\| = 0$, then $x_\ell = x_k, \forall \ell \geq k$. Thus, we already find a critical point of f. If we have in additionally that f is convex, then x_k is also a global solution of (OP).

2. If $\|x_k\| \neq 0$, for sufficiently small $\alpha_k > 0$, we have: $f(x_{k+1}) < f(x_k)$ (proof is an exercise). Thus, the objective function is monotonically decreasing along the iterations.

Useful practices concerning (GD) are:

1. *How to choose* x_0: very oftenly, we choose x_0 randomly (from a Gaussian or a uniform distribution).

2. *How to choose* α_k: The choice of α_k is a dedicated subject since the algorithm behaves differently, depending on the choice of α_k. In this lecture, we will only consider the most simple setting: $\alpha_k = \alpha$ is constant. Other criteria such as Armijo and Wolfe conditions were and will be discussed in the TP and TD.

3. *When to stop* (GD)*?* Many criteria are proposed in the literature:

- After reaching a fixed number of iterations (very common in deep learning practice).
- The gradient norm is sufficiently small: $\|\nabla f(x)\| \leq \epsilon$.
- The objective function does not decrease significantly enough $f(x_k) - f(x_{k+1}) \leq \epsilon$.

Example 2.1. Gradient descent for linear function: $f(x) = ax + b$. The update rule is given by:

$$x_{k+1} = x_k - \alpha_k a.$$

Thus, if α_k is constant, we have:

$$\lim_{k\to\infty} x_k = \begin{cases} -\infty & \alpha > 0 \\ x_0 & \alpha = 0 \\ +\infty & \alpha < 0 \end{cases},$$

which is logic since the function is not bounded below in case $\alpha \neq 0$.

Example 2.2. Gradient descent for quadratic loss function. Consider the scalar quadratic optimization problem:

$$\underset{x\in\mathbb{R}^d}{\text{Minimize}}\, f(x) := \frac{1}{2}x^\top \mathbf{A}x + b^\top x + c.$$

The gradient of x is given by:

$$\nabla f(x) = \mathbf{A}x + b.$$

Thus, the update rule for gradient descent with fixed step-size is given by:

$$x_{k+1} = x_k - \alpha_k(\mathbf{A}x_k + b) = (\mathbf{I} - \alpha_k\mathbf{A})x_k - \alpha_k b.$$

The limit of x_k will depend on the spectral of $\mathbf{A}$ and the choice of $\{\alpha_k\}_{k\in\mathbb{N}}$.

3 Theoretical guarantees of gradient descent algorithms

Well-known behavior of gradient descent is studied under the following assumption:

Definition 3.1 (L-smooth function)**.** A function is L-smooth if its gradient is L-Lipschitz, i.e., for all $x, y \in \mathbb{R}^d$, we have:

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2.$$

The following lemma provides an important estimation for L-smooth functions:

Proposition 3.2 (Upperbound of L-smooth functions)**.** *If f is L-smooth, then for all $x, y \in \mathbb{R}^d$, we have:*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|x - y\|_2^2.$$

Proof. Exercise. □

Lemma 3.3 (Descent lemma)**.** *Given a L-smooth function f and an iterate given by gradient descent with constant step-size $\alpha > 0$, we have:*

$$f(x_{k+1}) \leq f(x_k) - \left(\alpha - \frac{L\alpha^2}{2}\right)\|\nabla f(x_k)\|^2.$$

Moreover, if $\alpha = \frac{1}{L}$ and $C := \inf f > -\infty$, we have:

$$\min_{k=0,\dots,K} \|\nabla f(x_k)\|^2 \leq \frac{2L}{K}(f(x_0) - C) = O\left(\frac{1}{K}\right).$$

Proof. Using Proposition 3.2, we have:

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2 \\ &= f(x_k) - \alpha\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x_k)\|^2 \\ &= f(x_k) - \left(\alpha - \frac{L\alpha^2}{2}\right)\|\nabla f(x_k)\|^2. \end{aligned} \tag{1}$$

In case $\alpha = \frac{1}{L}$, the inequality (1) become:

$$f(x_k) - f(x_{k+1}) \leq \frac{1}{2L}\|\nabla f(x_k)\|^2. \tag{2}$$

By summing these inequalities for $k = 0, \ldots, K$, we obtain:

$$f(x_0) - C \geq f(x_0) - f(x_{K+1}) = \frac{1}{2L}\sum_{i=0}^{K}\|\nabla f(x_k)\|^2 \geq \frac{K+1}{2L}\min_{k=0,\ldots,K}\|\nabla f(x_k)\|^2,$$

which concludes the proof. □

Remark 3.4. From Lemma 3.3, we conclude that to find an ϵ critical point, i.e., $\|\nabla f(x)\| \leq \epsilon$, one needs $O(\frac{1}{\epsilon^2})$ iterations.

What happen if the function f is also convex In case f is convex, we can show guarantee on the objective function. It is given by the following result.

Proposition 3.5 (Gradient descent for convex, L-smooth functions)**.** *Suppose that $\alpha_k = \alpha = \frac{1}{L}$ and* (OP) *admit at least a global solution $x^\star$. The iterates x_k generated by gradient descent satisfy that:*

$$f(x_k) - f(x^\star) \leq \frac{L}{2k}\|\theta_0 - \theta^\star\|^2 = O\left(\frac{1}{k}\right).$$

Proof. Since f is convex, we have:

$$f(x^\star) \geq f(x) - \nabla f(x)^\top (x - x^\star).$$

Applying this inequality to (2) gives us:

$$\begin{aligned} f(x_k) &\leq f(x_{k-1}) - \frac{1}{2L}\|\nabla f(x_{k-1})\|^2 \\ &\leq f(x^\star) + \nabla f(x_{k-1})^\top (x_{k-1} - x^\star) - \frac{1}{2L}\|\nabla f(x_{k-1})\|^2 \\ &\leq f(x^\star) + \frac{L}{2}\left(\|x_{k-1} - x^\star\|^2 - \|\underbrace{x_{k-1} - \frac{1}{L}\nabla f(x_{k-1})}_{x_k} - x^\star\|^2\right) \end{aligned}$$

By summing these in inequality from $i = 0$ to k, we obtain :

$$\begin{aligned} kf(x^\star) + \frac{L}{2}\left(\|x_0 - x^\star\|^2 - \|x_k - x^\star\|^2\right) &\geq \sum_{i=1}^{k} f(x_i) \\ &\geq kf(x_k) \quad (\text{since } f(x_k) \text{ is monotone}), \end{aligned}$$

which conclude the proof. □

Remark 3.6. We see that with the convexity of f, we get a stronger result: we proved that the sequence $f(x_k) \to f(x^\star)$ when $k \to \infty$ with rate $O\left(\frac{1}{k}\right)$.

However, we are not unable to prove that $x_k \to x^\star$ (the convergence of iterates). Showing iterate convergence usually requires additional assumption such as Kurdyka-Łojasiewicz inequality. Without this type of hypothesis, the iterates convergence of gradient descent remains an open question.

What happen if the function f is μ-strongly convex

Definition 3.7 (μ-strongly convex function)**.** A function f is called μ-strongly convex if $f(\cdot) - \frac{\mu}{2}\|\cdot\|_2^2$ is convex.

When $\mu = 0$, strong convexity is equivalent to convexity.

Lemma 3.8 (Properties of strongly convex functions)**.** *Consider $f : \mathbb{R}^d \to \mathbb{R}$ a μ-strongly convex function, we have:*

1. $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{1}{2}\mu t(1-t)\|x-y\|^2, \forall x, y \in \mathbb{R}^d, \forall t \in [0,1]$.
2. *If f is C^1, then* $f(y) \geq f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|x-y\|^2, \forall x, y \in \mathbb{R}^d$.
3. $(\nabla f(x) - \nabla f(y))^\top (x-y) \geq \mu\|x-y\|^2, \forall x, y \in \mathbb{R}^d$.
4. *If f is C^2, then $\nabla^2 f(x) \succeq \mu \mathbf{I}$ (i.e., $\nabla^2 f(x) - \mu\mathbf{I}$ est positive semidefinite).*

Proof. Exercise. □

Proposition 3.9 (Gradient descent for μ-strongly convex, L-smooth functions)**.** *Assume the same hypothesis (on f and $\alpha_k = \alpha = \frac{1}{L}$) as in Proposition 3.5. In addition, we also suppose that f is mu-strongly convex ($\mu > 0$). Then, the iterates generated by* (GD) *verify:*

$$f(x_k) - f(x^\star) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^\star)), \forall k \in \mathbb{N}.$$

Proof. By applying the second Lemma 3.8 with $y = x^\star$, the optimal solution of f (whose existence and unicity are guaranteed by Lemma 3.8) gives :

$$\begin{aligned} f(x^\star) = f(y) &\geq f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2 \\ &\geq \inf_{y \in \mathbb{R}} f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2 \\ &= f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2. \end{aligned} \tag{3}$$

Combining this inequality with (1) gives:

$$\begin{aligned} f(x_{k+1}) - f(x^\star) &\leq f(x_k) - f(x^\star) - \frac{1}{2L}\|\nabla f(x_k)\|^2 \\ &\leq f(x_k) - f(x^\star) - \frac{\mu}{L}(f(x_k) - f(x^\star)) \\ &= \left(1 - \frac{\mu}{L}\right)(f(x_k) - f(x^\star))). \end{aligned}$$

The result is obtained by applying the previous inequality in a recursive manner. □

Remark 3.10. If we get $f(x_k) - f(x^\star) \leq c^k(f(x_0) - f(x^\star))$, we say that the objective function converge linearly.

Remark 3.11. Under the hypothesis of Proposition 3.9, we can also prove that $x_k \to x^\star$ linearly.

4 Summary on the theoretical guarantees of gradient descent

Hypothesis	Results	Convergence rate
• f is L-smooth. • $\inf_\theta f(\theta) > 0$. • Step-size $\alpha = \frac{1}{L}$	$\min_{k\in\mathbb{N}} \lVert\nabla f(\theta_k)\rVert$ tend vers zero	$O\left(\sqrt{\frac{1}{k}}\right)$
• f is L-smooth. • An optimal solution $\theta^\star$ exist. • Step-size $\alpha = \frac{1}{L}$ • f is convex.	$\lim_{k\to\infty} f(\theta_k) = f(\theta^\star)$	$O\left(\frac{1}{k}\right)$
• f is L-smooth. • Step-size $\alpha = \frac{1}{L}$ • f is μ-strongly convex.	• $\lim_{k\to\infty} \theta_k = \theta^\star$ • $\lim_{k\to\infty} f(\theta_k) = f(\theta^\star)$	$O\left(\left(1 - \frac{\mu}{L}\right)^k\right)$

Table 1: Résumé des résultats concernant la descente de gradient.

Lecture 4 : Lower-bound of first-order methods and Nesterov optimal algorithm

1 Introduction

In previous lectures, we already discussed about one of the most well-known optimization algorithms: *gradient descent* and its theoretical guarantee (on the convergence of iterates and of the objective functions). A natural question is: is gradient descent the *best* optimization that we can use when dealing with the class of L-smooth (and convex) functions? And what is the limit of an algorithm optimizing a function of this class? These questions will be (party) revealed in this lecture.

2 First-order methods and their fundamental limits

In this section, we consider the following question:

Question 2.1. Given an optimization algorithm $\mathcal{A}$ and a function class $\mathcal{F}$, investigate the worst objective gap at the kth iteration, i.e.,

$$\ell(\mathcal{F}, \mathcal{A}) := \sup_{f \in \mathcal{F}} f(x_k) - f^\star,$$

where x_k is the kth iteration generated by $\mathcal{A}$ and $f^\star = \inf_x f(x)$.

The sup operator in (2.1) implies that we would like find the *worst* function making the algorithm $\mathcal{A}$ sufferred the most. This gives us a lower-bound on the performance of the algorithm $\mathcal{A}$ when optimizing a function of f.

Of course, we expect that $\ell(\mathcal{F}, \mathcal{A})$ depends on the initialization x_0, the number of iterations k and the properties of the function class $\mathcal{F}$.

To this end, we consider $\mathcal{A}$ and $\mathcal{F}$ as follows:

1. $\mathcal{A}$ is a first-order algorithm, i.e., an iterative method whose sequence of iterates x_k satisfies:

$$x_k \in x_0 + \mathrm{Lin}\{\nabla f(x_0), \dots, \nabla f(x_{k-1})\}, k \geq 1.$$

In particular, gradient descent is a first-order method because:

$$\begin{aligned} x_k &= x_{k-1} - \alpha \nabla f(x_{k-1}) \\ &= x_{k-2} - \alpha \nabla f(x_{k-2}) - \alpha \nabla f(x_{k-1}) \\ &= \cdots \\ &= x_0 - \alpha(\nabla f(x_0) + \cdots + \nabla f(x_{k-1})). \end{aligned}$$

Attention: in general, first-order methods are much bigger since they refer to those that use only gradient.

2. $\mathcal{F}$ is the set of convex and L-smooth functions, i.e., f is convex and its gradient is L-Lipschitz.

Throughout this section, we will prove the following result:

Theorem 2.2 (Lower-bound of first-order methods performance)**.** *For any $x_0 \in \mathbb{R}^n$ and $1 \leq k \leq \frac{1}{2}(n-1)$, there exists a function $f \in \mathcal{F}$ such that for any first-order algorithm $\mathcal{A}$, we have:*

$$f(x_k) - f^\star \geq \frac{3L\|x_0 - x^\star\|^2}{32(k+1)^2}$$
$$\|x_k - x^\star\|^2 \geq \frac{1}{8}\|x_0 - x^\star\|^2.$$

where $x^\star \in \operatorname{argmin} f$ and $f^\star = f(x^\star)$.

The proof consists multiple steps, which are detailed below:

Construction of the functions f We consider a family of k quadratic functions ($k = 1, \ldots, n$) given by:

$$f_k(x) = \frac{L}{4}\left\{\frac{1}{2}\left[[x]_1^2 + \sum_{i=1}^{k-1}([x]_i - [x]_{i+1})^2 + [x]_k^2\right] - [x]_1\right\}$$

where $[x]_i$ is the ith coordinate of the vector x. We prove that $f_k \in \mathcal{F}$. Given a vector $s \in \mathbb{R}^n$, we have:

$$\begin{aligned} f_k(x+s) - f_k(x) =& \underbrace{\frac{L}{4}\left[[s]_1[x]_1 + \sum_{i=1}^{k-1}([s]_i - [s]_{i+1})^\top([x]_i - [x]_{i+1}) + [x]_k[s]_k - [s]_1\right]}_{\text{first-order term}} \\ &+ \frac{1}{2}\cdot\frac{L}{4}\underbrace{\left([s]_1^2 + \sum_{i=1}^{k-1}([s]_i - [s]_{i+1})^2 + [s]_k^2\right)}_{\text{second-order term}} \end{aligned}$$

Therefore, we conclude that:

$$0 \leq s^\top \nabla^2 f(x) s = \frac{L}{4}\left([s]_1^2 + \sum_{i=1}^{k-1}([s]_i - [s]_{i+1})^2 + [s]_k^2\right) \leq \frac{L}{4}\left([s]_1^2 + 2\sum_{i=1}^{k-1}([s]_i^2 + [s]_{i+1}^2) + [s]_k^2\right) \leq L\|s\|_2^2.$$

Therefore, $0 \preceq \nabla^2 f(x) \preceq L\mathbf{I}$. Therefore, f is convex and L-smooth (prove this is an exercise).

The optimal solution of the function f_k We compute the minimum of the function f_k. Since f is convex, it is sufficient to find x such that $\nabla f(x) = 0$. The gradient of f (up to a constant $\frac{L}{4}$) is given by:

$$\nabla f(x) = \begin{pmatrix} 2[x]_1 - [x]_2 \\ -[x]_1 + 2[x]_2 - [x]_3 \\ \ldots \\ -[x]_{k-2} + 2[x]_{k-1} - [x]_k \\ -[x]_{k-1} + 2[x]_k \end{pmatrix} - e_1$$

Therefore, to have $\nabla f(x) = 0$, one deduces (by induction) that:

$$[x]_i = (k - i + 1)[x]_k,$$

and hence, we obtain:

$$[x_k^\star]_i = \begin{cases} 1 - \frac{i}{k+1}, & i = 1, \dots, k \\ \mathbb{R}, & k+1 \le i \le n. \end{cases}$$

The optimal value of f_k is given by:

$$f_k^\star = f_k(x_k^\star) = -\frac{L}{8} \cdot \frac{k}{k+1} = -\frac{L}{8}\left(1 - \frac{1}{k+1}\right).$$

The iterates generated by first-order methods $\mathcal{A}$ Let's fix a value $1 \le p \le n$.

Proposition 2.3 (Properties of first-order methods)**.** *Let $x_0 = 0$. For any sequence $\{x_k\}_{k=0}^p$ generated by a first-order method $\mathcal{A}$, we have:*

$$x_k \in \mathbb{R}^{k,n} := \{x \in \mathbb{R}^n \mid x_i = 0, k+1 \le i \le n\}.$$

As a consequence, we have: $f_p(x_k) = f_k(x_k) \ge f_k^\star, \forall p = k, \dots, n$.

Proof. The proof is conducted by induction and the form of the gradient of f.

Since $f_p - f_k$ contains monomial of the form $x_j x_l$ where $j, l \ge k$, we have:

$$f_p(x_k) = f_k(x_k) \ge f_k^\star.$$

□

The actual proof of Theorem 2.2 WLOG, we consider first-order methods $\mathcal{A}$ whose initilization is $x_0 = 0$. Otherwise, we can use the following remark:

Remark 2.4. One can choose different $x_0 \neq 0$. However, the result remains the same by considering $g_k(\cdot) = f_k(\cdot + x_0)$.

We prove two claims one by one: since we assume $k \le \frac{1}{2}(n-1)$, we consider the function $f = f_{2k+1}$ and $x^\star = x_{2k+1}^\star$: We have:

$$\begin{aligned} \|x^\star - x_0\| &= \sum_{i=1}^{2k+1} \left(\frac{i}{2k+2}\right)^2 = \frac{1}{(2k+2)^2} \sum_{i=1}^{2k+1} i^2 \\ &= \frac{1}{(2k+2)^2} \frac{(2k+1)(2k+2)(4k+3)}{6} \le \frac{2k+2}{3}. \end{aligned}$$

1. **First claim**:

$$\frac{f(x_k) - f^\star}{\|x_0 - x^\star\|^2} \ge \frac{L}{8} \cdot \frac{\frac{1}{k+1} - \frac{1}{2k+2}}{\frac{2k+2}{3}} = \frac{3L}{32} \cdot \frac{1}{(k+1)^2}.$$

2. **Second claim**: Consider the quantity $\|x_k - x^\star\|^2$:

$$\begin{aligned} \|x_k - x^\star\|_2^2 &\ge \sum_{i=k+1}^{2k+1} ([x_{2k+1}^\star]_i)^2 = \sum_{i=k+1}^{2k+1} \left(1 - \frac{i}{2k+2}\right)^2 \\ &= k + 1 - \frac{1}{k+1} \sum_{i=k+1}^{2k+1} i + \frac{1}{4(k+1)^2} \sum_{i=k+1}^{2k+1} i^2 \\ &= k + 1 - \frac{3k+2}{2} + \frac{(2k+1)(7k+6)}{24(k+1)} \\ &\ge \frac{2k^2 + 7k + 6}{24(k+1)} \ge \frac{2k^2 + 7k + 6}{16(k+1)^2} \|x_0 - x^\star\|^2 \ge \frac{1}{8} \|x_0 - x^\star\|^2. \end{aligned}$$

3 Optimal algorithm with Nesterov acceleration

Knowing that the lower-bound is $O\left(\frac{1}{k^2}\right)$, we wonder if there exists an optimization attaining this limit. The answer is affirmative, which is the famous Nesterov acceleration method. It is defined as follows:

$$\begin{aligned} \lambda_k &= \frac{1+\sqrt{1+4\lambda_{k-1}^2}}{2} \\ \gamma_k &= \frac{1-\lambda_k}{\lambda_{k+1}} \\ x_{k+1} &= y_k - \frac{1}{L}\nabla f(y_k) \\ y_{k+1} &= (1-\gamma_k)x_{k+1} + \gamma_k x_k \end{aligned} \tag{Nes-Acc}$$

where $\lambda_0 = 0$ and $x_0 = y_0$ is a (random) initialization. The main result of this section is the following:

Theorem 3.1 (Theoretical guarantee of (Nes-Acc))**.** *Given a convex and L-smooth function f, the iterates generated by* (Nes-Acc) *satisfy:*

$$f(x_k) - f^\star \leq \frac{2L\|y_1 - x^\star\|^2}{k^2}.$$

where we assume $x^\star \in \operatorname{argmin} f$ is a non-empty set and $f^\star = f(x^\star)$.

Proof. Since f is convex and L-smooth, we have:

$$\begin{aligned} f\left(x - \frac{1}{L}\nabla f(x)\right) - f(y) &\leq f\left(x - \frac{1}{L}\nabla f(x)\right) - f(x) + \nabla f(x)^\top (x-y) \\ &\leq -\frac{1}{2L}\|\nabla f(x)\|^2 + \nabla f(x)^\top (x-y). \end{aligned}$$

Applying the previous results for $x = y_k$ and $y = x_k$, we have:

$$\begin{aligned} f(x_{k+1}) - f(x_k) &= f\left(y_k - \frac{1}{L}\nabla f(y_k)\right) - f(x_k) \\ &\leq -\frac{1}{2L}\|\nabla f(y_k)\|^2 + \nabla f(y_k)^\top (y_k - x_k) \\ &= -\frac{L}{2}\|x_{k+1} - y_k\|^2 - L(x_{k+1} - y_k)^\top (y_k - x_k). \end{aligned} \tag{1}$$

Similarly, we apply the result for $x = y_k$ and $y = x^\star$:

$$f(x_{k+1}) - f(x^\star) \leq -\frac{L}{2}\|x_{k+1} - y_k\|^2 - L(x_{k+1} - y_k)^\top (y_k - x^\star) \tag{2}$$

Multiplying (1) by $(\lambda_k - 1)$ and adding to (2) yields:

$$\lambda_k \underbrace{\left(f(x_{k+1}) - f^\star\right)}_{\delta_{k+1}} - (\lambda_k - 1)\underbrace{\left(f(x_k) - f^\star\right)}_{\delta_k} \leq -\frac{L\lambda_k}{2}\|x_{k+1} - y_k\|^2 - L(x_{k+1} - y_k)^\top (\lambda_k y_k - (\lambda_k - 1)x_k - x^\star)$$

By definition, we have: $\lambda_{k-1}^2 = \lambda_k^2 - \lambda_k$. Therefore, multiplying the previous inequality by λ_k, we have:

$$\begin{aligned} \lambda_k^2 \delta_{k+1} - \lambda_{k-1}^2 \delta_k &\leq -\frac{L}{2}\left(\|\lambda_k(x_{k+1} - y_k)\|^2 + 2\lambda_k (x_{k+1} - y_k)^\top (\lambda_k y_k - (\lambda_k - 1)x_k - x^\star)\right) \\ &= -\frac{L}{2}\left(\|\lambda_k x_{k+1} - (\lambda_k - 1)x_k - x^\star\|^2 - \|\lambda_k y_k - (\lambda_k - 1)x_k - x^\star\|^2\right) \\ &= -\frac{L}{2}\left(\|\lambda_{k+1} y_{k+1} - (\lambda_{k+1} - 1)x_{k+1} - x^\star\|^2 - \|\lambda_k y_k - (\lambda_k - 1)x_k - x^\star\|^2\right) \end{aligned}$$

Telescoping, we get:

$$\lambda_{k-1}^2 \delta_k \leq \frac{L}{2} \|y_1 - x^\star\|^2.$$

The proof is finished by noticing that $\lambda_{k-1} \geq \frac{k}{2}, \forall k \geq 1$. $\square$

4 Summary on the theoretical guarantees of first-order methods

Hypothesis	Gradient descent	Nesterov acceleration
• f is L-smooth. • An optimal solution $\theta^\star$ exist. • Step-size $\alpha = \frac{1}{L}$ • f is convex.	$f(x_k) - f^\star = O\left(\frac{1}{k}\right)$ $x_k \to x^\star \in \operatorname{argmin} f$?	$f(x_k) - f^\star = O\left(\frac{1}{k^2}\right)$ $x_k \to x^\star$

Table 1: Comparison between gradient descent and Nesterov acceleration

Lecture 5 : Stochastic gradient descent and theoretical properties

So far, we witnessed two first-order algorithms: gradient descent and its Nesterov acceleration. The common points of these algorithms is their access to an oracle of first-order derivatives (i.e., the gradient). In reality, such an oracle can be expensive, and one only has access to a noisy version of the true gradient. Stochastic gradient descent emerge as an alternative for its deterministic counterpart in those cases, and it is the main topic of this lecture.

1 Definition of stochastic gradient descent

Motivation: Adaptation of gradient descent to massive datasets Remind that in machine learning, many optimisation problem admits the following form:

$$\underset{\theta \in \mathbb{R}^d}{\text{Minimize}} \quad \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \underbrace{\ell(f(\theta, x_i), y_i)}_{\ell_i(\theta)}, \tag{1}$$

where $\mathcal{D} = (x_i, y_i)_{i=1}^n \in (\mathcal{X} \times \mathcal{Y})^n$ is a dataset, $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ is a loss function and $f(\theta, \cdot)$ is a parameterized model (for example, a linear model or a neural network).

If $n \gg 1$ (for example, $n \geq 10^6$), the estimation of gradient of $\mathcal{L}$ is expensive because:

$$\nabla \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \nabla \ell_i(\theta).$$

Different from the update rule of the original gradient descent for minimizning $\mathcal{L}$, which is given by:

$$\theta_{k+1} = \theta_k - \alpha_k \nabla \mathcal{L}(\theta_k), \tag{GD}$$

the stochastic version uses the following update rule:

$$\theta_{k+1} = \theta_k - \alpha_k \nabla \ell_i(\theta_k), \tag{SGD}$$

where $i \in \{1, \ldots, n\}$ is a randomly chosen index (usually from $\mathcal{U}(\{1, \ldots, n\})$ - the uniform distribution of the set $\{1, \ldots, n\}$). With this choise of i_k, even if (SGD) does not take the same direction as (GD) for each realisation of i_k, it does so in average because:

$$\begin{aligned}
\mathbb{E}[\theta_{k+1} \mid \theta_k] &= \theta_k - \alpha_k \mathbb{E}_i[\nabla \ell_i(\theta_k)] \\
&= \theta_k - \alpha_k \frac{1}{n} \sum_{i=1}^{n} \nabla_\theta \ell_i(\theta) \quad (\text{car } i_k \sim \mathcal{U}(\{1, \ldots, n\})) \\
&= \theta_k - \alpha_k \nabla \mathcal{L}(\theta_k).
\end{aligned}$$

Moreover, with the same computation cost, the stochastic gradient descent can do $O(n)$ steps, instead of only one (as in the gradient descent case), which makes this method very fast and competitive in practice.

Other forms of stochastic gradient descent More generally, the formula of stochastic gradient descent can be written as:

$$\theta_{k+1} = \theta_k - \alpha_k \underbrace{(\nabla \mathcal{L}(\theta_k) + \epsilon_k)}_{g_k(\theta_k)} \tag{SGD-G}$$

where ϵ_k is a pertubation term inserted to the kth iteration. Below are several popular choices of ϵ_k:

1. In the previous case, we have:

$$\epsilon_k \sim \mathcal{U}(\{\nabla \mathcal{L}(\theta_k) - \nabla \ell_i(\theta_k) \mid i = 1, \dots, n\})$$

2. Gaussian distribution: $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$.

In certain applications (for example, training with differential privacy), ϵ_k is sampled from Gaussian distribution rather than (SGD). As gradient descent, we establish the theoretical guarantees of the stochastic gradient descent in the following:

2 The convergence and the convergence rate of the stochastic gradient descent

In this section, we first introduce the important hypothesis for our theoretical results concerning the stochastic gradient descent defined in (SGD-G). After that, we treat three cases: $\mathcal{L}$ non convex, convex and μ-strongly convex, as we did with gradient descent.

2.1 The hypothesis on the stochastic gradient descent

Assumption 2.1 (Hypothesis on $\mathcal{L}$)**.** *The objective function $\mathcal{L}$ defined in Equation* (1) *is L-smooth.*

Assumption 2.1 can be satisfied when the function $\ell_i(\cdot), i = 1, \dots, n$ are L-smooth, which is usually the case for optimization problems such as linear regression or linear logistics.

Assumption 2.2 (Hypothesis on g_k)**.** *Assume that $g_k(\cdot), k \in \mathbb{N}$ verifies:*

1. $\mathbb{E}[g_k(\theta)] = \nabla \mathcal{L}(\theta)$, $\forall \theta \in \mathbb{R}^d$.

2. $\mathbb{E}[\|g_k(\theta) - \nabla \mathcal{L}(\theta)\|^2] \leq \sigma^2, \forall \theta \in \mathbb{R}^d$.

Several examples that verify Assumption 2.2 will be found in the tutorial session.

Le lemme suivant est la version stochastique du lemme de descente que nous avons vue dans le cours précédent.

Lemma 2.1 (Stochastic version of descent lemma)**.** *Consider $\mathcal{L}$ a function verifying Assumption 2.1 and g_k the stochatic gradient of $\mathcal{L}$ verifying Assumption 2.2, we have:*

$$\mathbb{E}[\mathcal{L}(\theta_{k+1}) \mid \theta_k] \leq \mathcal{L}(\theta_k) - \left(\alpha_k - \frac{L\alpha_k^2}{2}\right) \|\nabla \mathcal{L}(\theta_k)\|^2 + \frac{L\alpha_k^2}{2}\sigma^2.$$

Proof. By the property of an L-smooth function, we have:

$$\begin{aligned}\mathcal{L}(\theta_{k+1}) &\leq \mathcal{L}(\theta_k) + \nabla\mathcal{L}(\theta_k)^\top(\theta_{k+1} - \theta_k) + \frac{L}{2}\|\theta_{k+1} - \theta_k\|^2 \\ &\leq \mathcal{L}(\theta_k) - \alpha_k\langle\nabla\mathcal{L}(\theta_k), g_k(\theta_k)\rangle + \frac{L\alpha_k^2}{2}\|g_k(\theta_k)\|^2.\end{aligned}$$

Taking the expectation of the above inequality, we obtain:

$$\begin{aligned}\mathbb{E}[\mathcal{L}(\theta_{k+1}) \mid \theta_k] &\leq \mathcal{L}(\theta_k) - \alpha_k\|\nabla\mathcal{L}(\theta_k)\|^2 + \frac{L\alpha_k^2}{2}\mathbb{E}[\|g_k(\theta_k)\|^2 \mid \theta_k] \\ &\leq \mathcal{L}(\theta_k) - \alpha_k\|\nabla\mathcal{L}(\theta_k)\|^2 + \frac{L\alpha_k^2}{2}\left(\|\nabla\mathcal{L}(\theta_k)\|^2 + \mathtt{Var}[g_k(\theta_k) \mid \theta_k]\right) \\ &= \mathcal{L}(\theta_k) - \alpha_k\|\nabla\mathcal{L}(\theta_k)\|^2 + \frac{L\alpha_k^2}{2}\left(\|\nabla\mathcal{L}(\theta_k)\|^2 + \sigma^2\right) \quad \square\end{aligned}$$

2.2 Le cas non convexe

Theorem 2.2 (Stochastic gradient descent for L-smooth function)**.** *Suppose that $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ et $g_k : \mathbb{R}^d \to \mathbb{R}^d$ verifies Assumption 2.1 and Assumption 2.2, respectively. We suppose in addition that $C := \inf_{\theta\in\mathbb{R}^d}\mathcal{L}(\theta) > -\infty$ and $\alpha_k = \alpha < \frac{2}{L}$ is a constant. The iterations generated by* (SGD-G) *verify:*

$$\min_{k=0,\dots,K-1}\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] \leq \frac{\mathcal{L}(\theta_0) - C}{K(\alpha - \frac{L\alpha^2}{2})} + \frac{L\alpha}{1 - \frac{L\alpha}{2}}\sigma^2.$$

Proof. By taking the expectation of the inequality of Lemma 2.1, we obtain:

$$\mathbb{E}[\mathcal{L}(\theta_{k+1})] \leq \mathbb{E}[\mathcal{L}(\theta_k)] - \left(\alpha - \frac{L\alpha^2}{2}\right)\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] + \frac{L\alpha^2}{2}\sigma^2$$

By summing this inequality from $k = 0, \dots, K-1$, we have:

$$\mathbb{E}[\mathcal{L}(\theta_K)] \leq \mathbb{E}[\mathcal{L}(\theta_0)] - \left(\alpha - \frac{L\alpha^2}{2}\right)\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] + K\frac{L\alpha^2}{2}\sigma^2.$$

This implies:

$$\begin{aligned}\min_{k=0,\dots,K-1}\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] &\leq \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] \\ &\leq \frac{1}{K\left(\alpha - \frac{L\alpha^2}{2}\right)}\left(\mathbb{E}[\mathcal{L}(\theta_0)] - \mathbb{E}[\mathcal{L}(\theta_K)] + K\frac{L\alpha^2}{2}\sigma^2\right) \\ &\leq \frac{1}{K\left(\alpha - \frac{L\alpha^2}{2}\right)}\left(\mathbb{E}[\mathcal{L}(\theta_0)] - C + K\frac{L\alpha^2}{2}\sigma^2\right) \quad \square\end{aligned}$$

Remark 2.3. Si nous prenons $\alpha = \frac{1}{L\sqrt{K}}$, nous aurons $\mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|^2] \leq O\left(\frac{1}{\sqrt{K}}\right)$.

2.3 The strongly convex case

Remind that if $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is μ-strongly convex, then there eixsts an unique optimal solution $\theta^\star$ and for all θ, we have:

$$\mathcal{L}(\theta) - \mathcal{L}(\theta^\star) \leq \frac{1}{2\mu}\|\nabla\mathcal{L}(\theta)\|^2.$$

Theorem 2.4 (Stochastic gradient descent for μ-strongly convex and L-smooth functions)**.** *Suppose the same hypothesis as in Theorem 2.4. In addition, we suppose that f is μ-strongly convex ($\mu > 0$) and $0 < \alpha \leq \frac{1}{L}$. Then the iterates generated by* (SGD-G) *verify:*

$$\mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}^\star] \leq (1-\mu\alpha)^k \mathbb{E}[\mathcal{L}(\theta_0) - \mathcal{L}^\star] + \frac{L\alpha}{2\mu}\sigma^2, \quad \forall k \in \mathbb{N}.$$

Proof. Denote $\mathcal{L}^\star$ the minimum value of $\mathcal{L}$. By Lemma 2.1, we have :

$$\begin{aligned}
\mathbb{E}[\mathcal{L}(\theta_{k+1}) - \mathcal{L}^\star \mid \theta_k] &\leq \mathcal{L}(\theta_k) - \mathcal{L}^\star - \left(\alpha - \frac{L\alpha^2}{2}\right)\|\nabla\mathcal{L}(\theta_k)\|^2 + \frac{L\alpha^2}{2}\sigma^2 \\
&\leq \mathcal{L}(\theta_k) - \mathcal{L}^\star - 2\mu\left(\alpha - \frac{L\alpha^2}{2}\right)(\mathcal{L}(\theta_k) - \mathcal{L}^\star) + \frac{L\alpha^2}{2}\sigma^2 \\
&\leq \left(1 - 2\mu\alpha\left(1 - \frac{L\alpha}{2}\right)\right)(\mathcal{L}(\theta_k) - \mathcal{L}^\star) + \frac{L\alpha^2}{2}\sigma^2 \\
&\leq (1-\mu\alpha)(\mathcal{L}(\theta_k) - \mathcal{L}^\star) + \frac{L\alpha^2}{2}\sigma^2 \quad (\text{car } \alpha \leq \frac{1}{L})
\end{aligned}$$

By taking the expectation w.r.t. θ_k, we obtain:

$$\mathbb{E}[\mathcal{L}(\theta_{k+1}) - \mathcal{L}^\star] \leq (1-\mu\alpha)\mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}^\star] + \frac{L^2\alpha^2}{2}\sigma^2.$$

By recurrence, we can show that:

$$\mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}^\star] \leq (1-\mu\alpha)^k \mathbb{E}[\mathcal{L}(\theta_0) - \mathcal{L}^\star] + \frac{L\alpha}{2\mu}\sigma^2.$$

□

Remark 2.5. If the number of iterations k is fixed and sufficiently large, we can take:

$$\alpha = \frac{\ln k}{\mu k} \leq \frac{1}{L}.$$

Therefore, the bound Theorem 2.4 becomes:

$$\begin{aligned}
\mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}^\star] &= \left(1 - \frac{\ln k}{k}\right)^k \mathbb{E}[\mathcal{L}(\theta_0) - \mathcal{L}^\star] + \frac{L\ln k}{2k}\sigma^2 \\
&\leq \frac{1}{k}\mathbb{E}[\mathcal{L}(\theta_0) - \mathcal{L}^\star] + \frac{L\ln k}{2k}\sigma^2 \\
&= \tilde{O}\left(\frac{1}{k}\right).
\end{aligned}$$

3 Summary of theoretical guarantees of gradient descent and stochastic gradient descent

Hypothesis	Result	Convergence rate
• $\mathcal{L}$ is L-smooth. • g_k is not biased and of bounded variance. • $\inf_\theta \mathcal{L}(\theta) > 0$.	$\lim_{k\to\infty} \min_{k\in\mathbb{N}} \mathbb{E}[\|\nabla\mathcal{L}(\theta_k)\|] = 0$	$O\left(\sqrt[4]{\frac{1}{k}}\right)$
• $\mathcal{L}$ is L-smooth. • g_k is not biased and of bounded variance. • $\inf_\theta \mathcal{L}(\theta) > 0$. • An optimal solution exists. • $\mathcal{L}$ is convex.	$\lim_{k\to\infty} \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^\star)] = 0$	$O\left(\sqrt{\frac{1}{k}}\right)$
• $\mathcal{L}$ is L-smooth. • g_k is not biased and of bounded variance. • $\mathcal{L}$ is μ-strongly convex.	$\lim_{k\to\infty} \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^\star)] = 0$	$\tilde{O}\left(\frac{1}{k}\right)$

Table 1: Summary of stochastic gradient descent.

Hypothèses	Descente de gradient	Descente de gradient stochastique
$\mathcal{L}$ is L-smooth	$O\left(\sqrt{\frac{1}{k}}\right)$	$O\left(\sqrt[4]{\frac{1}{k}}\right)$
$\mathcal{L}$ is L-smooth and convex	$O\left(\frac{1}{k}\right)$	$O\left(\sqrt{\frac{1}{k}}\right)$
$\mathcal{L}$ is L-smooth and μ-strongly convex	$O\left(\left(1-\frac{\mu}{L}\right)^k\right)$	$\tilde{O}\left(\frac{1}{k}\right)$

Table 2: Comparison between gradient descent and stochastic gradient descent.

Lecture 6 : Adaptive methods - Adagrad and variants

In this lecture, we investigate adaptive methods, a family of optimization algorithms that become popular recently. Different from gradient descent or Nesterov acceleration iterations where all coefficients use the same step-size, algorithms of this family uses different step-sizes for different coordinates and

1 Adaptive methods - Adatative Gradien Descent

2 Motivations

Here are two examples of motivations:

1. **Quadratic optimization with gradient descent** : Consider an instance of quadratic functions $f : \mathbb{R}^2 \to \mathbb{R} : \theta \mapsto \frac{1}{2}\theta^\top \mathbf{A}\theta$ where:

$$\mathbf{A} = \begin{pmatrix} \mu & 0 \\ 0 & L \end{pmatrix}, L \geq \mu > 0.$$

Because f is μ-strongly convex and $\nabla f(\theta) = \mathbf{A}\theta$, f admits a unique optimal solution $\theta^\star = \mathbf{0}$. Moreover, we can explicitly calculate the update of gradient descent as:

$$\theta_k = \begin{pmatrix} 1-\mu\alpha & 0 \\ 0 & 1-L\alpha \end{pmatrix} \theta_{k-1} = \ldots = \begin{pmatrix} (1-\mu\alpha)^k & 0 \\ 0 & (1-L\alpha)^k \end{pmatrix} \theta_0$$

We observe that the convergence rate of two coefficients are different: $(1-\alpha\mu)^k$ and $(1-\alpha L)^k$.

In the regime where $\alpha\mu \approx 0$, the progress of the first coefficient is very slow because to reach ϵ, we will need:

$$k \approx \frac{1}{\alpha\mu} \log \frac{1}{\epsilon}.$$

A natural idea is to increase α because it will increase $\alpha\mu$ and reduce the number of necessary iterations to reach ϵ error. However, there is a limit that we can increase α: $\alpha \geq \frac{2}{L}$. Otherwise, the second coefficient will diverge because $|1-\alpha L| > 1$.

Another possible solution is to choose distinct steps $(\alpha_1, \ldots, \alpha_d)$ for each coefficient. In our example, if we choose $\alpha_1 = \frac{1}{\mu}$ and $\alpha_2 = \frac{1}{L}$, we arrive at the optimal solution in one step.

2. **(Still) quadratic case and stochastic gradient descent**: The second motivation comes from the optimization problems in machine learning: Imagin that you have a linear regression of the following form:

$$\underset{\theta\in\mathbb{R}^d}{\text{Minimize}} \quad \frac{1}{n}\|\mathbf{X}\theta - y\|^2 = \frac{1}{n}\sum_{\ell=1}^{n}(\theta^\top x_i - y_i)^2.$$

   Suppose that there are important features but it rarely appears (for example, being millionaire, ambidextrous, etc.), the stochastic gradient descent risk not to update the coefficients sufficiently regular because:

$$\theta_{k+1} = \theta_k - \alpha x_{i_k}(\theta_k^\top x_{i_k} - y_{i_k})$$

   and most of the time, the coefficients of x_{i_k} corresponding to the important features are nulls.

   The idea of adaptativity appears once again as an interesting solution. By using larger step-size from these coefficients, we can accelerate the convergence.

Nevertheless, this idea raise new problems: a priori, it is not trivial to choose good step-sizes for a specific optimization problem. This question pushed the following innovative idea: instead of choosing a good step-size from the beginnning, we use adaptive step-sizes that evolve through the iterations and incorporate (gradient) information. In the following section, we look at such an algorithm: Adaptive Gradient Descent (AdaGrad).

2.1 Adaptive Gradient Descent - AdaGrad

Consider the following generic optimization algorithm (OP),

$$\underset{\theta}{\text{Minimize}} \quad f(\theta) \tag{OP}$$

AdaGrad applies the following update rule:

$$\begin{aligned} g_{k+1} &= g_k + \nabla f(\theta_k) \odot \nabla f(\theta_k) \\ \theta_{k+1} &= \theta_k - \alpha_k \texttt{diag}(g_{k+1})^{-1/2}\nabla f(\theta_k), \end{aligned} \tag{1}$$

where $g_0 = \mathbf{0}$. By induction, we can prove that:

$$[g_{k+1}]_\ell = \sum_{i=0}^{k}([\nabla f(\theta_i)]_\ell)^2.$$

If this quantity is small, i.e., the ℓth coordinate is rarely updated, Adagrad will increase its step-size by dividing $\sqrt{[g_{k+1}]_\ell}$.

There are, in fact, other variants of Adagrad, with different norms, given by:

$$\begin{aligned} g_{k+1} &= g_k + \nabla f(\theta_k)^p \\ \theta_{k+1} &= \theta_k - \alpha_k \texttt{diag}(g_{k+1})^{-1/p}\nabla f(\theta_k). \end{aligned} \tag{2}$$

For $p = 2$, (2) reduces to (1). It is noteworthy that Adagrad does not belong to the set of first-order methods explained in the previous lecture because:

$$\theta_k \notin x_0 + \text{Lin}\{\nabla f(\theta_0), \nabla f(\theta_1), \ldots, \nabla f(\theta_{k-1})\}.$$

It is still a first-order method, though!!! (in a much larger sense).

2.2 Theoretical guarantee for AdaGrad in the minimization of convex functions

We prove a theoretical guarantee for Adagrad. In general, the performance of AdaGrad is not always better than that of gradient descent (with or without acceleration) from theoretical viewpoint. However, the following result implies that AdaGrad will perform much better when the gradients are sparse, which is a regular situation in reinforcement learning. In addition, the numerical result of plenty of models in deep learning suggests that the adaptive methods can greatly accelerate the training.

Theorem 2.1 (Convergence of AdaGrad for convex functions)**.** *Consider f a convex function which admits at least one optimal solution $\theta^\star$. Assume that $\|\theta_k - \theta^\star\|_\infty \leq D_\infty, \forall k \in \mathbb{N}$ and $\alpha_k = \alpha := \frac{D_\infty}{\sqrt{2}}$, we have:*

$$f(\bar{\theta}_{K+1}) - f(\theta^\star) \leq \frac{1}{K+1}\left(\alpha + \frac{1}{2\alpha}D_\infty^2\right)\sum_{\ell=1}^{d}\|g_{0:K}^\ell\|_2$$

where $\bar{\theta}_K = \frac{1}{K+1}\sum_{k=1}^{K+1}\theta_k$ and $g_{0:k}^\ell = \begin{pmatrix}[\nabla f(\theta_0)]_\ell & \dots [\nabla f(\theta_k)]_\ell\end{pmatrix}^\top$.

Proof. Denote $G_k = \texttt{diag}(g_k)^{1/2}$. By (1), we have:

$$\begin{aligned}\theta_{k+1} - \theta^\star &= \theta_k - \theta^\star - \alpha G_{k+1}^{-1}\nabla f(\theta_k)\\ G_{k+1}(\theta_{k+1} - \theta^\star) &= G_{k+1}(\theta_k - \theta^\star) - \alpha\nabla f(\theta_k)\end{aligned}$$

By taking the scalar product of the two above equations, we obtain:

$$(\underbrace{\theta_{k+1} - \theta^\star}_{\delta_{k+1}})^\top G_{k+1}(\theta_{k+1} - \theta^\star) = (\underbrace{\theta_k - \theta^\star}_{\delta_k})^\top G_{k+1}(\theta_k - \theta^\star) - 2\alpha\nabla f(\theta_k)^\top(\theta_k - \theta^\star) + \alpha^2\nabla f(\theta_k)^\top G_{k+1}^{-1}\nabla f(\theta_k).$$

By consequence, we have:

$$\nabla f(\theta_k)^\top\delta_k = \frac{\alpha}{2}\nabla f(\theta_k)^\top G_{k+1}^{-1}\nabla f(\theta_k) + \frac{1}{2\alpha}\left(\delta_k^\top G_{k+1}\delta_k - \delta_{k+1}^\top G_{k+1}\delta_{k+1}\right).$$

By summing this inequality for $k = 0, \dots, K$, we obtain:

$$\begin{aligned}\sum_{k=0}^{K}\nabla f(\theta_k)^\top\delta_k &= \frac{\alpha}{2}\sum_{k=0}^{K}\nabla f(\theta_k)^\top G_{k+1}^{-1}\nabla f(\theta_k) + \frac{1}{2\alpha}\sum_{k=0}^{K}\left(\delta_k^\top G_k\delta_k - \delta_{k+1}^\top G_{k+1}\delta_{k+1}\right)\\ &\quad + \frac{1}{2\alpha}\sum_{k=0}^{K}\delta_k^\top(G_{k+1} - G_k)\delta_k\\ &\leq \frac{\alpha}{2}\sum_{k=0}^{K}\nabla f(\theta_k)^\top G_{k+1}^{-1}\nabla f(\theta_k) + \frac{1}{2\alpha}\sum_{k=0}^{K}\delta_k^\top(G_{k+1} - G_k)\delta_k.\end{aligned}$$

since $g_0 = 0$. We analyse two terms:

$$\sum_{k=0}^{K}\nabla f(\theta_k)^\top G_{k+1}^{-1}\nabla f(\theta^k) = \sum_{k=0}^{K}\sum_{\ell=1}^{d}\frac{[\|\nabla f(\theta_k)\|]_\ell^2}{\|g_{0:k}^\ell\|_2} = \sum_{\ell=1}^{d}\sum_{k=0}^{K}\frac{[\|\nabla f(\theta_k)\|]_\ell^2}{\|g_{0:k}^\ell\|_2} \leq 2\sum_{\ell=1}^{d}\|g_{0:K}^\ell\|_2.$$

Indeed, it is sufficient to prove the following inequality: if we denote $\nu = \begin{pmatrix}\nu_0 & \dots \nu_d\end{pmatrix}^\top$ et $\nu_{0:\ell} = \begin{pmatrix}\nu_0 & \dots \nu_\ell\end{pmatrix}^\top$, then:

$$\sum_{i=0}^{d}\frac{\nu_i^2}{\|\nu_{0:t}\|_2} \leq 2\|\nu\|_2.$$

The proof can be done by induction. It is sufficient to show that:

$$2\|\nu_{0:d-1}\|_2 + \frac{\nu_d^2}{\|\nu\|} \leq 2\|\nu\|,$$

which is correct because for all $\beta \leq \alpha, \beta > 0$, we have:

$$2\sqrt{\beta^2 - \alpha^2} + \frac{\alpha^2}{\beta} \leq 2\beta.$$

We provide an upperbound for the second term as:

$$\sum_{k=0}^{K} \delta_k^\top (G_{k+1} - G_k)\delta_k \leq \sum_{k=0}^{K} D_\infty^2 \texttt{Trace}(G_{k+1} - G_k) \leq D_\infty^2 \texttt{Trace}(G_{k+1}) = D_\infty^2 \sum_{\ell=1}^{d} \|g_{0:K}^\ell\|_2$$

By combining the two upperbound and subtituting $\alpha = \frac{D_\infty}{\sqrt{2}}$, we get:

$$\sum_{k=0}^{K} \nabla f(\theta_k)^\top \delta_k \leq \left(\alpha + \frac{1}{2\alpha} D_\infty^2\right) \sum_{\ell=1}^{d} \|g_{0:K}^\ell\|_2 = \sqrt{2} D_\infty^2 \sum_{\ell=1}^{d} \|g_{0:K}^\ell\|_2.$$

The proof is finished by using the two following inequalities:

$$\nabla f(\theta_k)^\top \delta_k \geq f(\theta_k) - f(\theta^\star)$$
$$\frac{1}{K+1} \sum_{k=0}^{K} f(\theta_k) \geq f(\bar{\theta}_K)$$

since the function f is convex. □

3 Hybrid methods : RMSProp et Adam

Despite being one of the pioneer of the adaptive methods, AdaGrad is not the most frequently used algorithm in practice. In fact, for many learning tasks, first-order methods and hybrid methods that combine linear combination of past iterations and adaptive step-size are more preferable due to their excel on pratical performance. To concludethis course, we introduce two more hybrid methods: *Root Mean Square Propagation* (RMSProp) and *Adaptive Moment Estimation* (ADAM). We ignore their theoretical guarantees since the mathematical arguments are too technical and not intuitive. The theoretical guarantees for these methods constitute an active research domain until today.

RMSProp RMSProp uses two hyperparameters : $\alpha > 0$ (learning rate), $0 < \beta_2 < 1$ (parameters for the adaptive step-sizes)

$$\begin{aligned} v_{k+1} &= \beta_2 v_k + (1-\beta_2)[\nabla f(\theta_k) \odot \nabla f(\theta_k)] \\ \theta_{k+1} &= \theta_k - \alpha \cdot \texttt{diag}(\sqrt{v_{k+1}} + \epsilon)^{-1} \nabla f(\theta_k) \end{aligned} \tag{3}$$

ADAM ADAM uses three hyperparameters: $\alpha > 0$, and $\beta_1, \beta_2 \in (0,1)$.

$$\begin{aligned} v_{k+1} &= \beta_2 v_k + (1-\beta_2)[\nabla f(\theta_k) \odot \nabla f(\theta_k)] \\ m_{k+1} &= \beta_1 m_k + (1-\beta_1)\nabla f(\theta_k) \\ \widehat{v_{k+1}} &= \frac{1}{1-\beta_2^{k+1}} v_{k+1} \\ \widehat{m_{k+1}} &= \frac{1}{1-\beta_1^{k+1}} m_{k+1} \\ \theta_{k+1} &= \theta_k - \alpha \cdot \texttt{diag}(\sqrt{\widehat{v_{k+1}}} + \epsilon)^{-1} \widehat{m_{k+1}} \end{aligned} \tag{4}$$

The factors $1 - \beta_2^{k+1}$ and $1 - \beta_1^{k+1}$ are bias correction in the sum.

In practice, ADAM is used more than RMSProp and is crucial in the training of Large Language Models (LLMs).

Lecture 7 : Second-order methods - Newton algorithm

This lecture is dedicated to the second-order methods, notably the Newton algorithms and its theoretical guarantee. In general, these methods can enjoy a much faster convergence rate than first-order counterparts. Nevertheless, the cost per iteration makes them very expensive, hence hindering their usage in large-scale optimization. Adapting second-order methods to real-world scale problems is an active research domain, and might enable us to handle challenging optimization problems.

1 Second-order methods - the Newton step

So far, all algorithms that we have seen in this course only exploit first-order information, i.e., their update rules only use gradients. It is, thus, very tempting to exploit higher-order derivatives to see whether we can break the fundamental limit of first-order methods. Newton method, possibly one of the most well-known second-order methods, will be our focus of today lecture.

Given a C^2 function, the Newton update is given by:

$$x_{k+1} = x_k - \alpha \nabla^2 f(x_k)^{-1} \nabla f(x_k). \tag{Newton}$$

When $\alpha = 1$, it is called the *pure Newton step*. If the step-size $\alpha \neq 1$, then this is a damped Newton step.

There are, in fact, many interpretations and fun facts about (Newton):

Original Newton method is to find zero of functions, and not an optimization algorithm The original Newton method was proposed to find the x such that $f(x) = 0$ of a differentiable function $f : \mathbb{R} \to \mathbb{R}$. Starting from a guess point, the algorithm updates its iterations as:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

This rule is proposed due to the first-order approximation: $f(x + \delta) = f(x) + f'(x)\delta$. The pure (Newton) is when we apply the original Newton method for $f : \mathbb{R}^n \to \mathbb{R}^n : x \mapsto \nabla f(x)$, or more precisely:

$$\nabla^2 f(x_k)(x_{k+1} - x_k) - \nabla f(x_k) = 0.$$

Pure Newton step is the steepest descent method of a special metric In fact, the descent direction of gradient descent can be view as the minimizer of the following constrained problem:

$$\begin{aligned} \underset{p\in\mathbb{R}^d}{\text{Minimze}} \quad & \nabla f(x_k)^\top p \\ \text{s.t.} \quad & \|p\|_2 \le 1. \end{aligned}$$

The Newton step has the same interpertation, but with $\|p\|_f = p^\top \nabla^2 f(x_k) p$, i.e.,

$$\begin{aligned} \underset{p\in\mathbb{R}^d}{\text{Minimze}} \quad & \nabla f(x_k)^\top p \\ \text{s.t.} \quad & \|p\|_f \le 1. \end{aligned}$$

We can get the optimal direction by considering $\nabla^2 f^{1/2} p$.

Pure Newton step is the minimization of the surrogate of the function f In fact, the gradient descent step can be viewed as the minimizer of the following surrogate function:

$$f_{\text{GD}}(x) = f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2\alpha}\|x - x_k\|^2$$

where the minimizer is given by:

$$x^\star = x_k - \alpha \nabla f(x_k) = \text{gradient descent step.}$$

For Newton step, the surrogate function is the second order Taylor approximation of f, i.e.:

$$f_{\text{Newton}}(x) = f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top \nabla^2 f(x_k)(x - x_k),$$

given that $\nabla^2 f(x_k) \succ 0$.

Affine invariant of Newton step An important of (Newton) is that it is independent of linear or affine changes of coordinates. Suppose that we have a non-singular matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, and define $g = f \circ \mathbf{A}$, we have:

$$\nabla g(x) = \mathbf{A}^\top \nabla f(\mathbf{A}x) \qquad , \qquad \nabla^2 g(x) = \mathbf{A}^\top \nabla^2 f(\mathbf{A}x)\mathbf{A}.$$

Therefore, the Newton update of the function g at the point $y = \mathbf{A}^{-1}x$ is:

$$\begin{aligned} \mathbf{A}^{-1}x - \left(\nabla^2 g(\mathbf{A}^{-1}x)\right)^{-1} \nabla g(\mathbf{A}^{-1}x) &= x - (\mathbf{A}^\top \nabla^2 f(x)\mathbf{A})^{-1}\mathbf{A}^\top \nabla f(x) \\ &= \mathbf{A}^{-1}x - \mathbf{A}^{-1}\nabla^2 f(x)^{-1}\nabla f(x) \\ &= \mathbf{A}^{-1}(x - \nabla^2 f(x)^{-1}\nabla f(x)). \end{aligned}$$

Thus, the Newton update of the function g is that of the function f, up to the inverse of non-singular linear transformation $\mathbf{A}$. This is the unique property of Newton methods, unlike other first-order methods.

2 Newton algorithm and theoretical guarantees

The Newton algorithm is given as below: in each iteration:

1. Compute the Newton step p_k as in (Newton) and decrement $\lambda^2 := \nabla f(x)^\top \nabla^2 f(x)^{-1} \nabla f(x)$.
2. Line search: find step-size α so that $f(x - \alpha p_k) \le f(x) - c\alpha\lambda^2$ (Armijo rule) by backtracking.
3. Update: $x_{k+1} = x_k - \alpha p_k$.

Theoretical guarantees We employ two assumptions

Assumption 2.1 (Strongly convex and smooth)**.** *The function f is m-strongly convex and M-smooth, i.e.,*

$$m\mathbf{I} \leq \nabla^2 f(x) \leq M\mathbf{I}, \forall x.$$

Assumption 2.2 (L-Lipschitz hessian)**.** *The Hessian of f is Lipschitz continuous with constant L, i.e.,*

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2, \forall x, y.$$

In the analysis, the Newton algorithm will go through two stage:

1. *Damped Newton phase*: In this stage, the value function will decrease steadily by a constant $\gamma > 0$ and the norm of $\|\nabla f(x)\| \leq \delta > 0$.
2. *Pure Newton phase*: In the second stage, the norm of the gradient becomes small and the line search will always pick $\alpha_k = 1$ and:

$$\frac{L}{2m}\|\nabla f(x_{k+1})\|_2 \leq \left(\frac{L}{2m}\|\nabla f(x_k)\|_2\right)^2.$$

This is also known as quadratic convergence.

Before entering the technicalities, remind us of two identities that will be used repeatedly throughout the proof:

$$\begin{aligned} f(y) &= f(x) + \int_0^1 \nabla f(x + t(y - x))(y - x)dt, \\ f(y) &= f(x) + \nabla f(x)^\top (y - x) + \int_0^1 (1 - t)(y - x)^\top \nabla^2 f(x + t(y - x))(y - x)dt. \end{aligned} \tag{1}$$

Damped Newton phase By M-smoothness of the function f, we have:

$$\begin{aligned} f(x_k - \alpha p_k) &\leq f(x_k) - \alpha \nabla f(x_k)^\top p_k + \alpha^2 \frac{M\|p_k\|^2}{2} \\ &\leq f(x_k) - \alpha\lambda_k^2 + \frac{M}{2m}\alpha^2\lambda_k^2. \end{aligned}$$

Therefore, if we choose $\alpha \leq \frac{m}{M}$, it will satisfy the exit condition of the line search for $c < \frac{1}{2}$ because:

$$f(x_k - \alpha p_k) \leq f(x_k) - \alpha\left(1 - \frac{M}{2m}\alpha\right)\lambda_k^2. \tag{2}$$

Therefore, our step-size α found by line-search will satisfy:

$$\alpha_k \geq \beta\frac{m}{M},$$

where $\beta > 0$ is a constant dependent on our line search algorithm. As such:

$$f(x_{k+1}) - f(x) \leq -\frac{1}{2}\alpha\lambda^2 \leq -\beta\frac{m}{2M}\lambda^2 \leq -\alpha\beta\eta^2\frac{m}{M^2}.$$

because $\|\nabla f(x)\|_2 < \eta$ and $\lambda^2 \geq (1/M)\|\nabla f(x)\|^2$.

Quadratically convergent phase We prove next that if:

$$\eta \leq \frac{m^2}{2L},$$

then the line search will choose unit step, i.e., $\alpha_k = 1$. Indeed, since f is L-smooth, we have:

$$\begin{aligned}
f(x_{k+1}) &= f(x_k - \underbrace{\nabla^2 f(x_k)^{-1}\nabla f(x_k)}_{p_k}) \\
&= f(x_k) - \nabla f(x_k)^\top p_k + \int_0^1 (1-t) p_k^\top \nabla^2 f(x_k - tp_k) p_k dt \\
&= f(x_k) - \frac{1}{2}\lambda_k^2 + \int_0^1 (1-t) p_k^\top (\nabla^2 f(x_k - tp_k) - \nabla^2 f(x_k)) p_k dt \\
&\leq f(x_k) - \frac{1}{2}\lambda_k^2 + \left\| \int_0^1 (1-t) p_k^\top (\nabla^2 f(x_k - tp_k) - \nabla^2 f(x_k)) p_k dt \right\|^2 \\
&\leq f(x_k) - \frac{1}{2}\lambda_k^2 + \frac{L}{6} \cdot \|p_k\|^3.
\end{aligned}$$

Because

$$\begin{aligned}
\lambda_k^2 &= \|\nabla^2 f(x_k)^{-\frac{1}{2}} \nabla f(x_k)\|_2^2 = \|\nabla^2 f(x_k)^{\frac{1}{2}} p_k\|_2^2 \geq m\|p_k\|^2, \\
\|\nabla f(x_k)\| &= \|\nabla^2 f(x_k) p_k\|_2 \geq m\|p_k\|,
\end{aligned}$$

we have:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2}\lambda_k^2 + \frac{L}{6m^2}\lambda_k^2 \|\nabla f(x_k)\| = f(x_k) - \frac{1}{2}\lambda_k^2 \left(1 - \frac{L\|\nabla f(x_k)\|}{3m^2}\right).$$

Therefore, if $\eta \leq \frac{m^2}{2L}$, we have $f(x_k - d_k) \leq f(x_k) - \frac{1}{4}\lambda_k^2$ and line search will choose $\alpha_k = 1$.

Next, we will argue that when $\|\nabla f(x_k)\| \leq \frac{m^2}{2L}$, $\|\nabla f(x_\ell)\|, \ell \geq k$ will be smaller than this amount forever. Indeed,

$$\begin{aligned}
\|\nabla f(x_{k+1})\|_2 &= \|\nabla f(x_k - \nabla^2 f(x_k)^{-1}\nabla f(x_k))\|_2 \\
&= \left\| \nabla f(x_k) - \int_0^1 \nabla^2 f(x - t\nabla^2 f(x_k)^{-1}\nabla f(x_k)) \nabla^2 f(x_k)^{-1} \nabla f(x_k) dt \right\|_2 \\
&= \left\| \int_0^1 \left(\nabla^2 f(x - t\nabla^2 f(x_k)^{-1}\nabla f(x_k)) - \nabla^2 f(x_k)\right) \nabla^2 f(x_k)^{-1} \nabla f(x_k) \right\|_2 \\
&\leq \int_0^1 \left\| \left(\nabla^2 f(x - t\nabla^2 f(x_k)^{-1}\nabla f(x_k)) - \nabla^2 f(x_k)\right) \nabla^2 f(x_k)^{-1} \nabla f(x_k) \right\|_2 \\
&\leq L \int_0^1 t \left\| \nabla^2 f(x_k)^{-1} \nabla f(x_k) \right\| \|\nabla^2 f(x_k)^{-1}\| \|\nabla f(x_k)\|_2 \\
&\leq \frac{L}{2m^2} \|\nabla f(x_k)\|_2^2 \leq \|\nabla f(x_k)\|_2.
\end{aligned}$$

Moreover, the above inequality also shows that:

$$\|\nabla f(x_k)\|_2 = O\left(c^{2^k}\right), 0 \leq c < 1,$$

which is known as quadratic convergence speed of the norm of the gradient. This is much faster than linear convergence rate, which is read as: $\|\nabla f(x_k)\|_2 = O(c^k), c \in [0, 1)$. Finally, to get the quadratic convergence for the distance $\|x_k - x^\star\|$, we have:

$$m\|x_k - x^\star\|_2^2 \leq \langle \nabla f(x_k) - \nabla f(x^\star), x_k - x^\star \rangle \leq \|\nabla f(x_k)\| \|x_k - x^\star\|.$$

Hence,

$$\|\nabla f(x_k)\| \geq m\|x_k - x^\star\|_2.$$

Note that when one enters the quadratically convergent phase, to reach an error ϵ, one needs roughly $\log_2 \log_2 \frac{1}{\epsilon}$. For machine precision error (approximately 10^{-16}), one practically needs at most 6 iterations. Therefore, the number of iterations of Newton method is upper-bounded by:

$$\frac{f(x_0) - f^\star}{\eta} + 6.$$

The Newton method is suitable for small to medium sized problems since it suffers from a couple of problems:

1. Inversing the Hessian is expensive for large-scaled optimization problems since it might cost up to $O(d^3)$ basic operations where d is the problem dimension. Without special structure of the Hessian, this complexity per iteration is prohibitive.

2. If the function f is not strongly convex, the Hessian is not invertible and renders the Newton method useless. There are a couple of workarounds such as:

   (a) We choose the direction p_k as:

   $$p_k \in \operatorname*{argmin}_{p \in \mathbb{R}^d} \|\nabla^2 f(x_k)p - \nabla f(x_k)\|_2.$$

   In case $\nabla^2 f(x_k)$ is invertible, then we recover the Newton method. Otherwise, one needs a use a least square solver to find d_k.

   (b) We add a pertubation to the Hessian matrix and compute the direction as:

   $$p_k = (\nabla^2 f(x_k) + \epsilon \mathbf{I})^{-1} \nabla f(x_k).$$

   where $\epsilon > 0$ is a hyper-parameter of the algorithm.

3. If the function is not (strongly) convex, then the Newton direction is not necessarily a descent direction, i.e.,

   $$\lambda^2(x_k) := \nabla f(x_k)^\top \nabla^2 f(x_k)^{-1} \nabla f(x_k) \geq 0,$$

   if $\nabla f(x_k)$ is an eigenvector of $\nabla^2 f(x_k)$ with a negative eigenvalue. Using a similar model to the Newton method, many authors replaced the exact hessian matrix by a semi-definite approximation $\mathbf{B}_k$. These methods are known as quasi-Newton methods and they update the iteration along the direction d defined as:

   $$p_k \in \operatorname*{argmin}_{p \in \mathbb{R}^d} f(x_k) + \nabla f(x_k)^\top p + \frac{1}{2} p^\top \mathbf{B}_k p.$$

   If one chooses $\mathbf{B}_k = \nabla^2 f(x_k)$, we recover the Newton method.

Lecture 8 : Nonsmooth Optimization - Subgradient Descent

This lecture focus on non-smooth optimization, i.e., minimizing (or maximizing) a non-differentiable function. The main challenge of nonsmooth optimization is that one no longer has access to the gradient and higher-order derivatives, thus, renders all algorithms that we studied from the beginning useless. There are, in general, two main approaches to tackle non-smoothness: subgradient and proximal operators. The former will be introduced in this lecture, while the latter will be the subject of the next class.

1 Non-smooth optimization - examples and challenges

In many real-world problems, our objective function is happened not to be smooth. Here are two examples:

Example 1.1 (Max of two functions)**.** A common situation in optimization related to manufacturing or scheduling tasks is:

$$\underset{x\in\mathbb{R}^d}{\text{Minimize}} \quad f(x) := \max(h_1(x), h_2(x)),$$

where $h_1, h_2 : \mathbb{R}^d \to \mathbb{R}$. Even if h_1 and h_2 are (infinitely) differentiable, their maximum is not necessarily differentiable. Take $h_1(x) = -h_2(x) = -x$ as an example.

Example 1.2. Assume that one has a dataset $\{(x_i, y_i) \mid i = 1, \dots, n\}, (x_i, y_i) \in \mathbb{R}^d \times \{\pm 1\}$ and $d \gg n$. We want to learn a vector $\theta \in \mathbb{R}^d$ such that $\theta^\top x_i \approx y_i$ and we want the vector to be sparse, i.e., it has many zero coefficients. The LASSO provides a convex formulation to perform such a task, by optimizing the following function:

$$\underset{\theta}{\text{Minimize}} \quad f(x) := \sum_{i=1}^{n} (\theta^\top x_i - y_i)^2 + \lambda \|\theta\|_1$$

where $\|\theta\|_1 = \sum_{\ell=1}^{d} |\theta_\ell|$ is the ℓ_1 norm of θ. The function f is not differentiable because $\|\cdot\|_1$ is not.

What happens when the function is not smooth? Perhaps the simplest non-smooth (but convex) functions that we can think of is $f(x) = |x|$ can tell us a lot about non-smooth optimization. In fact, f is differentiable everywhere except $x = 0$ because:

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ x, & \text{otherwise} \end{cases} \quad \Longrightarrow \quad \nabla f(x) = \begin{cases} -1, & \text{if } x < 0 \\ \text{not defined}, & \text{if } x = 0 \\ 1, & \text{otherwise} \end{cases}.$$

Therefore, we can still perform "gradient descent", as long as our current iteration does not sit at 0. With fixed step-size $\alpha > 0$, our update rule becomes:

$$x_{k+1} = x_k - \alpha \texttt{sign}(x_k).$$

Typically, the dynamic generated by the above equation is: first, x_k will move towards 0 - the minimizer of f. Then, the sequence $\{x_k\}_{k\in\mathbb{N}}$ will oscillate around 0, and typically form a cycle of two points. Therefore, different from the smooth case where we already analysis, the iterates and the ojective function do not even converge to $x^\star$ or $f^\star = \inf f$. This suggests that a different approach, or more broadly, mathemtical framework is required to tackle this new situation. For that reason, we will study the notion of *subgradient*, a counterpart of gradient for non-smooth function.

2 Subdifferentiability and subgradient of a convex function

2.1 Definition and properties

Definition 2.1. Let $f : \mathbb{R}^n \to \bar{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$. Define $\texttt{dom}\, f = \{x \in \mathbb{R}^n \mid f(x) < \infty\}$. An element $g \in \mathbb{R}^d$ is called a subgradient of f at $x \in \texttt{dom}\, f$ if:

$$f(y) \geq f(x) + \langle g, y - x\rangle, \forall y \in \mathbb{R}^n.$$

The set of subgradient of f at a point x is denoted as $\partial f(x) \subseteq \mathbb{R}^d$.

Example 2.2. Take $f(x) = |x|$. We have:

$$\partial f(x) = \begin{cases} -1, & \text{if } x < 0 \\ [-1, 1], & \text{if } x = 0 \\ 1, & \text{otherwise} \end{cases}.$$

Remark that in Example 2.2, $\partial f(x) = \{\nabla f(x)\}$ whereever f is differentiable. This is not a coincidence, due to the following result:

Theorem 2.3 (Subgradient at a differentiable point)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex. If f differentiable at x, then $\partial f(x) = \{\nabla f(x)\}$.*

Proof. To prove that $\{\nabla f(x)\} \subseteq \partial f(x)$, we use a very similar argument as when we proved that for a C^1 function, one have:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

This will be left as an exercise. The rest of this proof will be dedicated to the other direction.

Assume that $g \in \partial f(x)$. By definition, we have:

$$f(x + d) - f(x) \leq g^\top d, \forall d \in \mathbb{R}^n.$$

Since the function f is differentiable at x, we also have:

$$\lim_{d\to 0} \frac{f(x + d) - f(x) - \nabla f(x)^\top d}{\|d\|} = 0.$$

This equation implies that for any $\epsilon > 0$, there exists $\delta > 0$ such that $\forall d, \|d\| \leq \delta$, we have:

$$\left\| \frac{f(x + d) - f(x) - \nabla f(x)^\top d}{\|d\|} \right\| \leq \epsilon,$$

or more precisely,

$$f(x+d) - f(x) \leq \nabla f(x)^\top d + \epsilon \|d\|, \forall d \in \mathbb{R}^n, \|d\| \leq \delta.$$

As a consequence, we obtain:

$$(\nabla f(x) - g)^\top d \leq \epsilon \|d\|, \forall d \in \mathbb{R}^n, \|d\| \leq \delta.$$

If $\nabla f(x) - g \neq 0$, one can choose $\epsilon < \|\nabla f(x) - g\|$, and $d = \delta \frac{\nabla f(x) - g}{\|\nabla f(x) - g\|}$, and obtain the contradiction. □

Therefore, if our function is differentiable, its subgradient reduces to its gradient. Note that the converse of Theorem 2.3 is also true, albeit the proof is omitted due to its complication.

Theorem 2.4 (Converse theorem of Theorem 2.3)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex. If $\partial f(x) = \{\nabla f(x)\}$, then f differentiable at x.*

Next, we investigate the optimality condition for a non-smooth (but convex) function.

Theorem 2.5 (Optimality condition)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function. A solution $x \in \mathbb{R}^d$ is a minimizer of f if and only if $0 \in \partial f(x)$.*

Proof. If x is a minimizer of f, then $0 \in \partial f(x)$ by definition. For the converse direction, it is immediately from the definition of subgradient, which is:

$$f(y) \geq f(x) + 0^\top (y - x) = f(x), \forall y \in \mathbb{R}^d.$$

□

2.2 Calculus with subgradient

Subgradient calculus share many gradient calculus, except for the chain rule. We consider the subgradient of f under certain operators in the following:

Theorem 2.6 (Subgradient calculus)**.** *Let $f, g : \mathbb{R}^n \to \mathbb{R}$ and $f_1, \ldots, f_n : \mathbb{R}^n \to \mathbb{R}$ be convex functions. We have:*

$$\begin{aligned} \partial(\alpha f)(x) &= \alpha \partial f(x) \\ \partial(f+g)(x) &= \partial f(x) + \partial g(x) \\ \partial(f \circ (\mathbf{A}x + b)) &= \mathbf{A}^\top \partial f(\mathbf{A}x + b) \end{aligned}$$

These results (except the first) one are quite complicated to prove. We accept them without actually proving them.

3 Subgradient descent algorithm and its theoretical guarantees

In this section, we investigate the subgradient descent method and its theoretical guarantees. Similar to gradient descent, subgradient descent update the current iteration by following the opposite direction of its subgradient, i.e.,

$$x_{k+1} = x_k - \alpha_k g_k, \quad g_k \in \partial f(x_k). \tag{1}$$

If f is differentiable, we recover the gradient descent method (see Theorem 2.4). The main theoretical guarantee for subgradient descent is the following:

Theorem 3.1 (Theoretical guarantee for subgradient descent)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex. Assume that f admit at least one minimzer $x^\star$. We have:*

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{1}{2} \cdot \frac{\|x_0 - x^\star\|_2^2 + \sum_{i=1}^2 \alpha_k^2 \|g_k\|^2}{\sum_{k=0}^{K-1} \alpha_k}.$$

Proof. We have:

$$\begin{aligned}
\|x_{k+1} - x^\star\|^2 &= \|x_k - x^\star - \alpha_k g_k\|^2 \\
&= \|x_k - x^\star\|^2 - 2\alpha_k g_k^\top (x_k - x^\star) + \alpha_k^2 \|g_k\|^2 \\
&\leq \|x_k - x^\star\|^2 - 2\alpha_k (f(x_k) - f(x^\star)) + \alpha_k^2 \|g_k\|^2.
\end{aligned}$$

Telescoping these inequalities with $k = 0, \dots, K-1$, we got:

$$2 \sum_{k=0}^{K-1} \alpha_k (f(x_k) - f(x^\star)) \leq \|x_0 - x^\star\|^2 - \|x_K - x^\star\|^2 + \sum_{k=0}^{K-1} \alpha_k^2 \|g_k\|^2.$$

Since:

$$2 \sum_{k=0}^{K-1} \alpha_k (f(x_k) - f(x^\star)) \geq 2 \left(\sum_{k=0}^{K-1} \alpha_k \right) \min_{k=0,\dots,K-1} (f(x_k - f(x^\star))),$$

divding both sides with $2 \left(\sum_{k=0}^{K-1} \alpha_k \right)$ yields the result. □

Corollary 3.2 (Guarantees for bounded subgradient function)**.** *Assume the same hypothesis as in Theorem 3.1, and that there exists a constant $G >$ such that $\|g\| \leq G, \forall g \in \partial f(x), \forall x$, then:*

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{1}{2} \cdot \frac{\|x_0 - x^\star\|_2^2 + G^2 \sum_{k=0}^{K-1} \alpha_k^2}{\sum_{k=0}^{K-1} \alpha_k}.$$

We investigate several cases:

1. If $\alpha_k = \alpha > 0$, then we have:

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{1}{2} \cdot \left(\frac{\|x_0 - x^\star\|_2^2}{K\alpha} + \alpha G^2 \right).$$

2. If $\alpha_k = \frac{1}{k}$, then we have:

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{1}{2} \cdot \frac{\|x_0 - x^\star\|_2 + G^2 \pi/4}{\ln k}.$$

3. If $\alpha = \frac{1}{\sqrt{k}}$, then we have:

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{1}{2} \cdot \frac{\|x_0 - x^\star\|_2 + G^2 \ln k}{\sqrt{k}}.$$

4. Lower-bound of the upper-bound: Note that

$$\frac{\|x_0 - x^\star\|^2 + G^2 \sum_{k=0}^{K-1} \alpha_k^2}{\sum_{k=0}^{K-1} \alpha_k} \geq \frac{\|x_0 - x^\star\|^2}{\sum_{k=0}^{K-1} \alpha_k} + \frac{1}{k} \left(\sum_{k=0}^{K-1} \alpha_k \right) \geq 2 \cdot \frac{\|x_0 - x^\star\|^2}{\sqrt{k}} = O\left(\frac{1}{\sqrt{k}} \right).$$

Therefore, $O(1/\sqrt{k})$ is the best one can get if we follow this analysis.

Polyak's step-sizes With which step-size strategies can we achieve the optimal rate $O(1/\sqrt{k})$? One of them is known as Polyak's step-sizes. It will be described in the following:

Assume that we know the optimal value $f^\star = f(x^\star)$, we choose:

$$\alpha_k = \frac{f(x_k) - f(x^\star)}{\|g_k\|^2}.$$

Theorem 3.3 (Theoretical guarantee for Polyak's step-sizes)**.** *Assume the same hypothesis as in Theorem 3.1, and that there exists a constant $G >$ such that $\|g\| \leq G, \forall g \in \partial f(x), \forall x$, then:*

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{G\|x_0 - x^\star\|}{\sqrt{k}}.$$

Proof. We consider:

$$\begin{aligned}
\|x_{k+1} - x^\star\|^2 &= \|x_k - x^\star - \alpha_k g_k\|^2 \\
&= \|x_k - x^\star\|^2 - 2\alpha_k g_k^\top (x_k - x^\star) + \alpha_k^2 \|g_k\|^2 \\
&\leq \|x_k - x^\star\|^2 - 2\alpha_k (f(x_k) - f(x^\star)) + \alpha_k^2 \|g_k\|^2 \\
&= \|x_k - x^\star\|^2 - 2\alpha_k (f(x_k) - f(x^\star)) + \alpha_k^2 \|g_k\|^2 \\
&= \|x_k - x^\star\|^2 - \frac{(f(x_k) - f(x^\star))^2}{\|g_k\|^2} \\
&\leq \|x_k - x^\star\|^2 - \frac{(f(x_k) - f(x^\star))^2}{G^2}.
\end{aligned}$$

Therefore, telescoping and using the fact that $f(x_k) - f(x^\star) \geq \min_{\ell=0,\dots,K-1} f(x_\ell) - f(x^\star), \forall K > k$, we get:

$$\min_{k=0,\dots,K-1} f(x_k) - f(x^\star) \leq \frac{G\|x_0 - x^\star\|}{\sqrt{k}}.$$

□